Tile Drainage: State of Knowledge

1 Abstract

Artificial subsurface (tile) drainage is among the most consequential anthropogenic modifications of the field-to-stream pathway, draining roughly 12–17% of the world's cultivated land and more than 22 Mha in the United States alone (Valipour 2014; Valayamkunnath et al. 2020). This review consolidates the state of knowledge on drained-land hydrology, water quality, agronomy, and management, emphasising well-supported findings, mechanisms, and quantified magnitudes over open questions. Its unifying conclusion is that drainage is fundamentally a hydrological intervention: by intercepting water that would otherwise recharge groundwater or leave as surface runoff and routing it rapidly to streams, tiles commonly carry 40–65% of streamflow in humid settings, dominate field-scale water export, elevate low flows and baseflow, shorten travel times, and homogenise catchment response (King, Fausey, and Williams 2014; Schilling, Gassman, et al. 2019). This hydrology governs the water-quality outcomes that follow. Relative to undrained or surface-drained land, subsurface drainage reduces runoff, sediment, and phosphorus but opens an efficient conduit for nitrate, the dominant and best-quantified export; nitrate loads are hydrologically controlled, strongly seasonal, and concentrated in high-flow episodes, with concentrations routinely near or above the 10 mg L^{-1} drinking-water limit. The presumption of a filtering "clean" subsurface pathway has been overturned: a small, hydraulically active volume of macropores and cracks delivers particulate phosphorus and sediment to drains within hours (Pease, King, et al. 2018), so tiles frequently dominate edge-of-field phosphorus loads (Pease, King, et al. 2018; Sharpley 1995). Against these losses, a tiered hierarchy of in-field, drainage-design, and edge-of-field practices produces quantified but scale-attenuated reductions—cover crops clustering around 40–60% nitrate-load reduction (Malone, Obrycki, et al. 2018; Malone, Jaynes, Kaspar, et al. 2014; Malone, Radke, et al. 2023), alongside controlled drainage, bioreactors, wetlands, and saturated buffers governed by residence time, temperature, and loading (L. Christianson, R. Christianson, Helmers, et al. 2013; Jaynes, Kaspar, et al. 2008)—so meeting downstream targets requires stacking complementary measures. Drainage also reliably raises yields, reclaims saline and waterlogged land, and returns favourable economic and social benefits, while validated process models, geophysics, and remote sensing underpin design and mapping. Because independent methods, climates, and scales converge on the same mechanisms and comparable magnitudes, these conclusions can be stated with confidence, and recognising drainage water quality as a design criterion is their consolidated implication.

2 Introduction

Artificial subsurface (tile) drainage is among the most consequential anthropogenic modifications of the field-to-stream pathway in agricultural landscapes. By lowering the water table and routing excess soil water rapidly to outlets, drainage converts poorly drained and waterlogged land into some of the world's most productive and intensively farmed cropland. Its footprint is correspondingly large: roughly 12–17% of cultivated land is artificially drained worldwide (Valipour 2014), about

three-quarters of surveyed United States producers report drained cropland (Morton et al. 2015), nearly 90% of Finnish farmers rate subsurface drainage as important—ahead of irrigation (Peltonen-Sainio, Sorvali, and Kaseva 2021), and remote-sensing inventories now place the drained United States extent at more than 22 Mha (Valayamkunnath et al. 2020). Because tile networks intercept water that would otherwise recharge shallow groundwater or leave as surface runoff and deliver it directly to streams, drainage behaves as a master variable of drained-land hydrology and water quality: in humid settings tiles commonly carry a large and measurable fraction of streamflow and dominate field water export (King, Fausey, and Williams 2014; Williams, King, and Penn 2018; Schilling, Gassman, et al. 2019). Understanding this engineered re-plumbing of the landscape is therefore central to managing both agricultural productivity and the quality of the waters that receive drainage.

The hydrologic changes wrought by drainage are double-edged. Relative to undrained or surface-drained land, improved subsurface drainage generally reduces surface runoff, peak flow, sediment, and particulate phosphorus, yet it opens an efficient conduit for nitrate, coupling tile-drained cropland to hypoxia in the Gulf of Mexico and to the eutrophication of inland waters (FAUSEY, BROWN, et al. 1995; SKAGGS, BREVE, and GILLIAM 1994; THOMAS et al. 1995; Gramlich et al. 2018; Raymond, David, and Saiers 2012). The once-common presumption that the soil profile filters subsurface water into a "clean" pathway has been overturned: a small volume of hydraulically active macropores delivers phosphorus, sediment, and conservative tracers to drains within hours of application or rainfall (Pease, King, et al. 2018). At the same time drainage remains a foundational production technology that raises yields on excess-water-stressed soils, reclaims saline and waterlogged land, and returns favourable financial and social benefits (N. Singh et al. 2022; Y. Yang, Ao, et al. 2025). Reconciling these competing outcomes—water-quality protection against agronomic and economic gain—is the defining management challenge of drained agriculture.

The knowledge synthesised here rests on several mutually reinforcing lines of evidence. Continuously monitored plots, fields, and watershed outlets quantify flows and loads; tracer, isotope, and geochemical studies partition flow pathways and fingerprint contaminant sources; controlled columns, tanks, and lysimeters isolate mechanisms; long-term records spanning decades reveal trends and legacies; and a mature toolkit of process-based and data-driven models, together with near-surface geophysics and remote sensing, simulates and locates buried networks (E. Smith and Kellman 2011a; A. Singh 2019; R. Skaggs, Youssef, and Chescheir 2012; Allred, Martinez, Fessehazion, Rouse, Williamson, et al. 2020). Where these independent approaches converge, findings can be stated with strong confidence, and it is that consolidated evidence—rather than open questions—that this review sets out to establish.

This paper is the state-of-knowledge component of a two-part synthesis. It consolidates what the evidence has established about agricultural subsurface drainage—the consolidated findings, mechanisms, and quantified outcomes that are well supported—and how strongly each is supported; the open questions, contested results, and emerging frontiers are treated in a companion frontier component. The organisation follows the causal logic of drained-land systems. Because hydrology is the master control on everything that follows, we begin with drainage flow pathways, the modified flow regime, and its scaling. We then turn to contaminant loss and its mechanisms—nitrate export and its biophysical controls; phosphorus and sediment loss and the preferential transport that drives it; and the further dimensions of pesticides, microbes, trace elements, and greenhouse gases. Management is treated next along the pathway from field to outlet: in-field nutrient and cropping practices, drainage design and water-table management, and edge-of-field treatment technologies with their watershed-scale integration. We then consolidate the agronomic, reclamation, and socio-economic benefits of drainage, and close with the modeling, measurement, and mapping toolkit on which the physical and management conclusions rest. Throughout, quantified magnitudes are

developed within their respective thematic sections; the aim of this synthesis is to bring those threads together into a coherent account of what is now known about tile drainage.

3 Drainage Hydrology: Flow Pathways, Regime, and Scaling

Subsurface drainage reshapes the field and catchment water balance in ways that are now well quantified across observational, tracer, and modelling lines of evidence. Tile networks intercept water that would otherwise recharge shallow groundwater or return as surface runoff and deliver it rapidly to streams, so drained landscapes carry a large and measurable share of streamflow, a distinctive flow regime, and a homogenised, scale-invariant response. This section establishes that physical hydrology—the flow pathways, the modified regime, and its scaling—on which the later water-quality and management findings depend.

3.1 Tile Drainage as a Dominant Flow Pathway

Watershed-scale studies consistently establish that tile drains route a large fraction of precipitation and streamflow. In an Ohio watershed, 21% of precipitation (206 mm) was recovered through tiles, accounting for 41% of annual and 47% of monthly discharge (King, Fausey, and Williams 2014), and equivalent-medium modelling of an Iowa watershed attributed 97 mm of tile drainage—13% of precipitation and 39% of streamflow (Thomas et al. 2016). In the Boone River, end-member mixing and SWAT bracketed the tile contribution at roughly 46–54% of discharge, with SWAT estimating about 53.7% (Boone) and 65.8% (Lyons Creek) of annual yield (Schilling, Gassman, et al. 2019); snowmelt-dominated modelling placed tile flow near 16% of annual precipitation (about 37% of yield), with tiling reducing surface runoff and increasing subsurface flow and yield (Rahman et al. 2014). Field-scale export can be even more tile-dominated: tile discharge accounted for $83 \pm 20\%$ of Ohio water export (Williams, King, and Penn 2018), and most discharge from an Indiana closed depression left through tiles rather than surface flow (Williams, Livingston, et al. 2019). Across humid settings the drain share commonly falls in the 40–65% range.

Partitioning between tile and surface pathways varies systematically with hydrologic condition and drainage area. In Iowa, tile drainage averaged 30% of streamflow during events but 61% during non-event periods; the event ratio Q_T/Q rose with drainage area while the non-event ratio fell, and annual Q_T/Q increased log-linearly from 0.29 at 6 km^2 to 0.39 at 45 km^2 ($R^2 = 0.98$) (Thomas et al. 2016; King, Fausey, and Williams 2014).

Tile contributions peak when soils are wet and evapotranspiration low. Ohio tile discharge was strongly seasonal (median monthly summer 0.9 mm versus winter 23.4 mm) (King, Fausey, and Williams 2014), Boone River drainage dominated flow from March through June (Schilling, Gassman, et al. 2019), and simulations indicated the largest baseflow effect from March to July—over 40% more baseflow in drained watersheds and about 60% more in April–June (Schilling and Helmers 2008). In cold regions tile flow tracks snowmelt, with a seasonal absence from May to September (Kompanizare et al. 2024), and water-table position relative to tile depth gates drainage: in a well-drained Iowa block the water table lay below tile depth 72% of the time, steepening gradients during wet periods (Logsdon and Cambardella 2020).

3.2 Modification of the Flow Regime

3.2.1 Low Flows and Baseflow

A robust, repeated finding is that tile drainage elevates low flows and the baseflow fraction. Iowa modelling showed drained landscapes carried far higher baseflow (150 mm, 72%, at 24 m spacing)

than undrained (52 mm, 25%), with drainage acting primarily on the baseflow portion of the hydrograph (Schilling and Helmers 2008). Ohio simulations found tiles greatly increased low flow while decreasing higher flows less, raised annual average daily flow about 5%, and lowered the probability of the water table entering the root zone from about 50% to about 5% (Rathore, Sloan, and Painter 2025). Routing analysis reinforced that tiles increase low flows, decrease intermediate flows, and minimally affect the largest flows (Sloan, Mantilla, et al. 2017). In Latvia, drainage raised mean summer–autumn runoff by 132–186% and winter low-water runoff by 53–286%, with annual mean, minimum, and maximum discharge up 46%, 158%, and 26% (Apsite et al. 2017). Channelized Ohio headwaters showed far fewer zero-discharge days (8 versus 87.5 d yr^{-1} in natural streams), inverting part of the accepted headwater paradigm (King, Smiley, and Fausey 2009).

3.2.2 Peak Flows and Flood Response

Effects on the largest peaks are consistently modest, but scale- and condition-dependent for smaller ones. Routing modelling concentrated peak-flow reductions at intermediate scales (100–1000 km^2), falling from 26% at the watershed to 11% at the basin outlet for one event, with negligible field-scale effect on the largest floods; placement mattered, with draining headwaters producing the greatest reduction (Sloan, Mantilla, et al. 2017). Field-scale Iowa scenarios showed no significant difference in mean annual peak discharge across six soil types, indicating drainage chiefly alters pathway rather than the largest peaks (Sloan, Basu, and Mantilla 2016). Ohio simulations found tiles could enhance event peaks for sub-threshold or dry-condition events but reduced them for larger events, cutting streamflow in 23 of the 25 largest events (Rathore, Sloan, and Painter 2025). Under expanded tiling, smaller peaks became more frequent while larger peaks declined, steeper rising limbs implying that reduced large-event peaks arose from reduced volumes rather than altered timing (Rahman et al. 2014). Distributed modelling similarly found drainage decreases peak flow while increasing recession volume, reproducing observed behaviour with only 21% of the area tiled (Fonley et al. 2021).

3.2.3 Flashiness, Drought, and Long-Term Trends

Flashiness responds to drainage in a soil- and slope-dependent way. A global meta-analysis of 464 watersheds found overall flashiness (Richards–Baker Index) similar regardless of drainage extent but diverging by setting: in flat watersheds (<2% slope) medium/high drainage raised the index, whereas in sloping watersheds low extent had the highest, with drainage extent correlating positively with agricultural land use and clay content and negatively with slope (S. Zhou and Margenot 2025). Field-scale Iowa simulations showed drainage decreased flashiness for low-conductivity soils and increased it for high-conductivity soils, homogenising response with an optimal spacing near 15 m (Sloan, Basu, and Mantilla 2016), while Ohio modelling found overall flashiness decreased with tiles, minimised near 50 m spacing (Rathore, Sloan, and Painter 2025). Across 30 Nordic–Baltic catchments, flashiness related negatively to baseflow index and positively to drained share and intensity, and computing it from hourly rather than daily discharge increased it nearly threefold (Deelstra et al. 2014a; Deelstra et al. 2014b). At the regime scale, drainage significantly increased streamflow-drought duration and intensity while reducing frequency, each 10% increase in tile-drained area lengthening drought duration by 0.03 d, raising intensity 12%, and lowering frequency 0.10 events yr^{-1} (Adelsperger et al. 2024). Indiana trend analysis found increasing low- and medium-flow trends with weakened precipitation linkage in heavily drained areas (S. Kumar, Merwade, et al. 2009), although a synthesis of seven Midwestern rivers concluded that inter-annual variability in streamflow and baseflow was controlled primarily by current- and previous-year pre-

cipitation (Baeumler and S. Gupta 2020).

3.3 Scale Dependence, Homogenization, and Travel Times

Artificial drainage homogenises catchment response and reduces the scale dependence of hydrologic signatures. Non-tiled Iowa watersheds show decreasing area-normalized peak discharge and increasing baseflow ratio with scale, whereas tiled basins behave scale-invariantly; within-basin variability in recession coefficients declines as drained fraction rises, and the tiled/non-tiled distinction disappears above roughly 2200 km^2 (Boland-Brien, Basu, and Schilling 2014). In Minnesota's Redwood basin, mean travel time fell 41% (77 to 45 d), year-to-year variability was reduced, and travel time became independent of catchment size above about 200 km^2 (Danesh-Yazdi et al. 2016), with streamflow dynamics becoming more linear after 1976 despite little change in precipitation (Foufoula-Georgiou et al. 2015). Isotope work in a UK catchment found short mean transit times (<1 yr, faster in artificial drains) and concluded that drainage homogenised distributed responses (Dimitrova-Petrova et al. 2020), and an Indiana watershed behaved as an effectively homogeneous unit describable by a single exponential travel-time distribution with a 2–3 d mean (Basu et al. 2010); even so, tile contributions remained detectable beyond 16,000 km^2 in the Boone River (Schilling, Gassman, et al. 2019).

Modelling and tracers quantify how drainage shortens groundwater travel times. In Iowa, raising drainage density shortened mean travel time and raised the tile contribution to baseflow to an upper bound of 37%, while increasing channel incision from 0.3 to 2.7 m lowered mean travel time near-linearly and raised the tile contribution from 30% to 54% (Schilling, Jindal, et al. 2012). In Bear Creek, mean groundwater travel time fell from 5.57 to 1.07 yr at high drainage intensity (versus 181.6 yr presettlement), cutting riparian-buffer residence time to 15.5–36.7 d (Schilling, Wolter, et al. 2015). Analytical solutions confirm that under tile drainage groundwater levels are lower and decline faster, total groundwater discharge to the river increases, and the tile contribution grows with density, though direct groundwater discharge during recession is reduced (X. Liang, Y. Zhang, and Schilling 2015); recession of heavily tiled watersheds is correspondingly more linear (Schilling and Helmers 2008).

Runoff generation is governed by pathway-activation thresholds. In an Indiana closed depression, rainfall alone related linearly to cumulative flow with no threshold, but a clear threshold emerged when antecedent soil-moisture deficit and rainfall were summed; tile flow generally began earlier, peaked earlier, and lasted longer than surface flow (Williams, Livingston, et al. 2019). Tile celerity there was controlled by antecedent wetness—preferential flow mattering more when dry and stored pre-event water dominating when wetter—with groundwater the primary tile source (mean transit time 245 d) (Williams and McAfee 2021). In near-level Canadian vertisols, overland flow generally activated before tile flow, and direct preferential flow into tiles was uncommon (Kokulan, Macrae, Ali, and Lobb 2019). Across 795 US–Canadian events, 43% produced both surface and tile flow and 57% tile flow only, surface flow contributing a median 12% among dual-pathway events (Macrae, Ali, et al. 2019); in Kansas tile-outlet terraces, baseflow contributed 38–51% of discharge (Stops et al. 2022).

Tracer and geochemical studies partition tile flow into fast preferential and slower matrix components whose balance shifts with season and wetness. In a French field, dry-condition rainwater transferred directly through macropores whereas saturated conditions were dominated by pre-event soil water; surface and tile drainage transferred 44% and 20% of annual effective rainfall, with 65% of infiltrating water not captured by tiles (Gaillot et al. 2023), and catchment monitoring found the fast preferential peak contributing about 15% of water and sediment fluxes (Grangeon et al. 2021). Ohio master-recession analysis apportioned annual tile discharge into 9% quick-old, 39%

quick-new, 42% slow-old, and 10% slow-new components (Nazari, Ford, and King 2021), preferential flow accounted for a higher fraction in clay than loam (Pluer et al. 2020), and Swedish work attributed autumn events to preferential flow and spring events to matrix flow (Salazar, Wesström, and Joel 2011). Flow is highly concentrated in time: in Ohio, surface runoff occurred 41 ± 42 and tile drainage 240 ± 69 d yr^{-1}, with 80% of tile discharge delivered in 21% of the year (Williams, King, and Penn 2018).

3.4 Repartitioning of Water Relative to Undrained Land

Relative to undrained or surface-drained conditions, improved subsurface drainage generally reduces surface runoff, peak flow, sediment, and phosphorus even as it increases nitrate delivery (FAUSEY, BROWN, et al. 1995; SKAGGS, BREVE, and GILLIAM 1994; THOMAS et al. 1995; Gramlich et al. 2018). In Oregon, drainage reduced watershed runoff and sediment yield by about 65% and 55%, eliminated baseflow, and lowered peak flow (ISTOK and KLING 1983); Albanian sites recorded surface-runoff reductions of 30–56% (Grazhdani, Jacquin, and Sulce 1996); and Minnesota modelling found reduced runoff, soil loss, and sediment yield, especially under no-till (Maalim and Melesse 2013). In North Carolina, improved drainage cut surface runoff by 192 mm yr^{-1} (66%) while raising total outflow by 10% and strongly attenuating peak runoff (101 to 28 mm day^{-1}) (KONYHA, SKAGGS, and GILLIAM 1992).

At the field-water-balance scale, the tile drain, not the soil surface, is usually the principal conduit in humid settings. In Indiana, surface runoff was only 1–10% of annual precipitation versus 26–45% for subsurface drainage (simulated balance 7% runoff, 34% drain flow, 51% evapotranspiration), with all ponding generated by saturation excess (Saadat et al. 2020). Iowa pothole modelling found tile discharge and evapotranspiration accounting for 52.4% and 47.5% of precipitation (Amado et al. 2018; Martin, Kaleita, and Soupir 2019), and UK field drains were the dominant pathway transferring runoff and most sediment to the stream, responding in every monitored event (Deasy et al. 2009). The direction of these shifts is texture-dependent: on sandy loams subsurface drainage minimised runoff and total-phosphorus loss whereas the reverse held on clay loam, phosphorus speciation being mainly a function of soil texture (Eastman et al. 2010). Mechanistically, good drainage increases infiltration and lowers solute concentrations in surface runoff (Tong et al. 2016; Tao, S. Wang, D. Xu, H. Yuan, et al. 2017), and by lowering the water table it limits saturation-excess runoff and surface sealing (Augeard et al. 2005).

4 Nitrate Export: Magnitudes and Biophysical Controls

Nitrate is the overwhelmingly dominant nitrogen species leaving artificially drained cropland, and its export is one of the best-quantified consequences of subsurface drainage. Across continuously monitored plots, drainage districts, and watershed outlets, three findings recur with strong support: the subsurface pathway carries the great majority of nitrogen loss, the magnitude of that loss is governed first by hydrology, and the underlying concentration–discharge behaviour reflects rapid, macropore-mediated delivery from largely organic-derived nitrate sources that are only partly attenuated by denitrification. This section establishes those magnitudes and biophysical controls—the "what and why" of nitrate export—and couples load to flow specifically for nitrate; broader flow-regime behaviour is treated in the hydrology section, and all management levers are deferred to the management and design sections.

4.1 Subsurface drainage as the dominant nitrogen pathway

4.1.1 Pathway dominance

Multiple independent studies establish that in tile-drained landscapes the subsurface pathway carries most of the nitrogen. In intensively monitored Norwegian catchments, subsurface drainage carried 76–91% of runoff and 87–95% of total-N (TN) loss, with drainage TN concentrations 2–4 times those in surface runoff and nitrate-N constituting 71–85% of TN (Bechmann 2014). New Zealand mole-pipe systems captured about 62% of surplus rainfall as drainage and routed roughly 80% of soluble N, while surface runoff dominated P, sediment, and *E. coli* (Monaghan, L. Smith, and Muirhead 2016). At watershed scale, tile drainage accounted for 56% of annual discharge and 44–82% (mean 62%) of NO_3-N export in Ohio (Williams, King, and Fausey 2015), averaged 57% of streamflow and dominated nitrate export in Indiana (Grose et al. 2022), and in Iowa the largest drains contributed 70–87% of exported water and nitrate (Coupe, Thornburg, et al. 2020). In flat Illinois terrain, subsurface stations lost 13–30 kg ha^{-1} yr^{-1} of nitrate-N (16–26% of applied N), with more than 70% of samples exceeding the 10 mg L^{-1} maximum contaminant level (MCL), while surface runoff was negligible (Dattamudi et al. 2020).

4.1.2 Loads, concentrations, and MCL exceedance

Reported magnitudes span a wide but coherent range. Field- and watershed-scale nitrate-N yields commonly reach or exceed 20–60 kg ha^{-1} yr^{-1} in maize–soybean systems (Gentry, David, Below, et al. 2009; Gentry, David, and McIsaac 2014; Dattamudi et al. 2020; Tomer, Moorman, and Rossi 2008), and tile concentrations frequently approach or exceed the 10 mg L^{-1} drinking-water limit. In three Iowa drainage districts tile water averaged about 13 mg L^{-1} NO_3-N and exceeded the standard 85–90% of the time (Schilling, Jones, et al. 2012), and across three catchments the flow-weighted mean was 17.1 mg L^{-1} with 74.2% of samples above the MCL (Ikenberry, Soupir, Schilling, et al. 2014). New Zealand tile drains exported 14–109 kg ha^{-1} yr^{-1} TN (58–90% nitrate) (Tanner and Sukias 2011); Illinois riverine and tile yields averaged 30–32 and ranged to 57–59 kg ha^{-1} yr^{-1} (Gentry, David, and McIsaac 2014); and very high concentrations characterise some European and irrigated systems, with Italian drainage averaging 94–216 mg NO_3 L^{-1} (ROSSI, CIAVATTA, and ANTISARI 1991) and Polish water usually exceeding 70 mg L^{-1} (Dragon et al. 2016). A key methodological finding is that interannual variability in load greatly exceeds that of concentration: across 29 Iowa watersheds, flow-weighted concentration (0.6–15.5 mg L^{-1}) was a more robust indicator of management change than load or yield (Danalatos et al. 2022). Early Iowa work already characterised nitrate loss as "highly variable and strongly dependent on the amount of water lost" (J. BAKER et al. 1975), and long-term records document real declines: German drainage N discharge fell from 33 to 10 kg N ha^{-1} yr^{-1} between the 1980s and 1990s, driven mainly by a reduced N surplus (Hirt, Hammann, and Meyer 2005).

4.1.3 Fertilizer input, hydrology, and soil-N legacy

At regional scale, fertilizer input and watershed hydrology jointly regulate export. About 34% of applied fertilizer N was exported to streams in the Mississippi basin, wetter watersheds exported a larger fraction, and tile drainage acted as a hydrologic modifier that increases throughput and shortens residence time, raising nitrate loss to the Gulf of Mexico (Raymond, David, and Saiers 2012). Direct comparison confirmed the causal link: nitrate load from an unfertilized Minnesota field was 58% of that from a fertilized field (Ghane, Ranaivoson, et al. 2016). Multivariate models combining water yield, prior corn yield, and fertilizer sales explained 96% of annual nitrate-yield

variation (Gentry, David, and McIsaac 2014). Losses stem partly from asynchrony between mineralization and crop uptake and from large mineralizable soil-N pools, so land-management effects may only become visible after years to decades owing to soil-N legacies (Cambardella et al. 1999; Bauwe et al. 2020).

4.2 Hydrologic control of nitrate export and seasonality

The single most robust cross-scale finding is that nitrate load is controlled by drainage discharge. Annual nitrate-N loss relates strongly to annual flow, with differences among drain spacings explained by flow (Kladivko and Bowling 2021; Bauwe et al. 2020); in Italy NO_3 loss correlated with drainage water ($r = 0.93$) but only weakly with precipitation ($r = 0.45$) (ROSSI, CIAVATTA, and ANTISARI 1991). Because export tracks flow, interannual variability is driven principally by precipitation and drainage volume rather than by concentration (Danalatos et al. 2022; J. BAKER et al. 1975).

High flows dominate export. More than half of Iowa catchment nitrate export occurred during the highest 10% of daily flows (Ikenberry, Soupir, Schilling, et al. 2014), and 79–95% of Indiana NO_3-N export occurred at the 60th–100th flow percentile (Speir et al. 2021). Because loads track flow they are strongly seasonal, concentrating in the non-growing season when evapotranspiration is low and drainage is high: about 70% of annual drainage and nitrate load occurred in the November–April fallow season in Indiana (Kladivko and Bowling 2021; Kladivko, Frankenberger, et al. 2004), and 81% of Ohio tile discharge occurred in the nongrowing season, so nitrate and TN loads were greater then despite higher growing-season concentrations under corn (King, Williams, and Fausey 2016). Short episodes can dominate: 21 kg N ha^{-1} was lost in only 11 days during snowmelt in Japan (Hayashi and Hatano 1999). The broader flow-regime behaviour underlying this coupling is treated in the hydrology section.

4.3 Concentration–discharge behaviour, sources, and denitrification

4.3.1 Concentration–discharge behaviour and rapid response

High-frequency records reveal characteristic concentration–discharge (C–Q) signatures. Annual nitrate export is largely chemostatic—yields rise roughly proportionately with runoff—and persistent chemostasis has been documented in Illinois and Ohio tile drainage (Speir et al. 2021; Hu et al. 2024; Williams, King, and Fausey 2015). Storm-event hysteresis is predominantly counterclockwise (77% of Indiana storms; $> 75\%$ of a long-term Ohio record), indicating dominance of distal nitrate sources during delivery (Speir et al. 2021; W. Liu, S. Tian, et al. 2022). Analysis concluded that intensive subsurface drainage, rather than deep groundwater, was the dominant factor shaping nitrate C–Q relationships (W. Liu, S. Tian, et al. 2022). Tile nitrate responds within hours of water inputs: after irrigation it rose sharply within hours then dropped abruptly, with preferential flow intercepted by the drain originating close to and above the tile (Mohanty, Bowman, Hendrickx, Simunek, et al. 1998), and bypass (macropore) flow carried most leached N in Japan (Hayashi and Hatano 1999). Threshold behaviour is common: tile discharge began once the soil-moisture deficit exceeded -17 ± 3 mm and the water table rose above 0.84 ± 0.03 m (Williams, Penn, and McAfee 2022), and antecedent moisture modulated discharge and solute pulses (Trentman et al. 2020; Kennedy et al. 2012).

4.3.2 Sources, transformation, and denitrification

Dual-isotope studies indicate that drainage nitrate is often derived largely from organic sources (manure and soil organic matter) via nitrification, with denitrification not always dominant. In Ontario, δ^{15}N and δ^{18}O relationships lacked the 2:1 denitrification slope, pointing to organic sources and continued nitrification through all seasons (E. Smith and Kellman 2011a; E. Smith and Kellman 2011b), and rapid nitrification of applied ammonium (> 90% within 72 h) supplies ready nitrate for leaching (Hu et al. 2024). Where denitrification is active it is detectable: along Polish flow paths nitrate declined while sulfate, excess N_2, δ^{15}N, and δ^{18}O increased (Dragon et al. 2016), and nested French watersheds implied 34–82% nitrate retention dominated by underground denitrification (Billy, Birgand, Sebilo, et al. 2011; Billy, Birgand, Ansart, et al. 2013). In Iowa drainageways, denitrification in slowly permeable soils could remove essentially all upland groundwater nitrate, functioning as a natural treatment system (Schilling, Tomer, et al. 2007; Cambardella et al. 1999). Yet where drains intercept water before it reaches these attenuating zones, export is high: tile discharge is dominated by pre-event water—three-component separation gave 65% groundwater and 29% shallow soil water—yet tile drains carried the highest NO_3-N of any source and contributed over 99% of watershed nitrate export (Williams, Penn, and McAfee 2022; Grose et al. 2022).

5 Phosphorus and Sediment Loss and Preferential Transport

5.1 Overview

Phosphorus (P) moves in far smaller mass than nitrogen, yet even at low concentrations it drives eutrophication, and the long-standing presumption that the soil profile filters subsurface water into a "clean" P pathway has been overturned. A geographically diverse body of monitoring, isotopic and radionuclide tracing, soil micromorphology, dye and smoke-injection experiments, and process modelling now supports a well-consolidated picture. Subsurface drainage typically captures the largest share of field water and, despite lower P concentrations than surface runoff, frequently dominates edge-of-field P *loads*, often as particulate P (PP) delivered by preferential flow (Pease, King, et al. 2018; Van Esbroeck et al. 2016; Uusitalo, Turtola, Kauppila, et al. 2001; Laubel et al. 1999). Dissolved P release is tightly coupled to soil P status, degree of P saturation, and subsoil sorption, with legacy P sustaining loads at watershed scale (Sharpley 1995; Beauchemin et al. 2003; Jarvie et al. 2017). A tiny, hydraulically active macropore volume—earthworm burrows, cracks, root channels, and installation scars—governs rapid solute and sediment delivery and reorganises soil structure over years to decades (Shipitalo, Nuutinen, and Butt 2004; Villholth, Jensen, and Fredericia 1998; Montagne, Cornu, Le Forestier, Hardy, et al. 2008). Losses are highly concentrated in time, dominated by the non-growing season and a few large events (Van Esbroeck et al. 2016; Brendel et al. 2019). This section synthesises these established magnitudes and mechanisms; P-removal technologies and the P responses of controlled drainage are treated in the edge-of-field and design sections, respectively.

5.2 Phosphorus Loss Pathways and Water Partitioning

Tile drainage commonly captures the largest share of field water export: subsurface discharge accounted for $81 \pm 23\%$ of annual discharge in the Eastern Corn Belt (Pease, King, et al. 2018) and 78–90% in Ontario (Van Esbroeck et al. 2016), and installing drainage re-routes water from surface- to drainage-dominated flow (Turtola and Paajanen 1995; BENGTSON et al. 1995). A recurring distinction is that surface runoff usually carries higher P *concentrations* whereas tile drainage—carrying more water—can dominate annual *loads*. In Ohio, flow-weighted DRP and

TP concentrations were greater in surface runoff, yet subsurface discharge supplied 71% of DRP and 69% of TP load (Pease, King, et al. 2018); the same contrast appears in long-term Swedish monitoring (Norberg et al. 2022). Tiles exported 40–77% of TP and 19–67% of DRP load in Ontario (Van Esbroeck et al. 2016), 21–52% of TP in Wisconsin (Madison et al. 2014), and 65.4% of annual TP at a southern-Ontario field (H. Qi et al. 2018). Models corroborate the partitioning, attributing 66% of DRP and 76% of TP loss to subsurface drainage against observed 70% and 80% (Grewal et al. 2025; Sadhukhan, Z. Qi, T. Zhang, Tan, and L. Ma 2019).

Tile dominance is not universal. In snowmelt-affected clay landscapes of the northern Great Plains, overland flow contributed 72–89% of runoff and $>$ 95% of SRP and TP because tiles were decoupled from the surface by frozen ground (Kokulan, Macrae, Lobb, et al. 2019; Kokulan, Macrae, Ali, Lobb, et al. 2021), and surface runoff also prevailed under some tillage, pasture, and flooded-paddy systems (S. Zhao et al. 2001; Melland et al. 2008; Darzi-Naftchali, Shahnazari, and Karandish 2016).

5.3 Particulate versus Dissolved Phosphorus and Bioavailability

A central finding is that PP is frequently the dominant form in tile drainage, contrary to the earlier presumption that the profile filters subsurface water to dissolved-dominated. In Finnish clays PP was the grand-mean 92% of TP in both runoff and drainage (Uusitalo, Turtola, Kauppila, et al. 2001); at Rothamsted's Broadbalk drains PP averaged 54% of TP and was strongly related to particulate matter (J. Lu 2003; J. Lu 2004); and PP dominated catchment losses in Denmark and elsewhere (Grant et al. 1996; Brendel et al. 2019; Salm et al. 2012). Dissolved forms nonetheless dominate in snowmelt-driven, grassland, high-STP, and organic-soil systems: tile SRP averaged 60% of TP across Michigan sites (Clement and Steinman 2017), and DRP dominated fertilised continuous-grass losses (T. Zhang, Tan, Zheng, and Drury 2015). Scottish work showed molybdate-reactive P was only 42% of subsurface TP loss, underscoring the need to measure all P forms (Hooda et al. 1999).

Because algal availability, not total mass, governs eutrophication risk, bioavailable-P estimates matter: desorbable PP and DRP were of equal importance to algal-available P in Finnish clays (Uusitalo, Turtola, Kauppila, et al. 2001; Uusitalo, Turtola, Puustinen, et al. 2003), and only 8–18% of grassland lysimeter TP was molybdate-reactive (Bilotta et al. 2008).

5.4 Soil Controls on Phosphorus Release and Legacy Phosphorus

Dissolved P loss is tightly coupled to soil P status. Loss in surface runoff and drainage related strongly to soil P content (r^2 = 0.85–0.96 for surface, 0.86 for subsurface) (Sharpley 1995), and higher soil-test P (STP) raised DRP in both tiles and runoff (Osterholz, Hanrahan, and King 2020). Threshold behaviour is common: a change point in STP marks where extractable P and DRP rise more steeply (McDowell and Sharpley 2001), and rainfall sharply increased subsurface DRP where STP exceeded 41 mg kg^{-1} (Ni, Y. Yuan, and W. Liu 2020). Where soils sorb P weakly, dissolved losses rise: A-horizon degree of P saturation (DPS) alone explained 83% of tile TP variance, rising to 95–96% with water-extractable P and sorption grouping (Beauchemin et al. 2003), and a DPS threshold near 0.25 predicted sandy-soil SRP (Siemens et al. 2004). The highest-leaching profiles combined large labile topsoil P pools with low-sorption subsoils, so subsoil characteristics must be included in risk assessment (J. Liu et al. 2024). Subsurface P losses are generally small in fine-textured soils at moderate STP but become significant in sandy, organic, and very high-P soils from long-term fertilisation or manuring (Sims, Simard, and Joern 1998). Sequential fractionation links pools to loss, with P lost to water associated with short-term ($<$ 24 h) exchangeable pools

tied to Al, Fe, and Ca (McDowell, Sinaj, et al. 2001).

Legacy P is manifest at watershed scale: cumulative SRP loads in Lake Erie tributaries rose 94–253% after 2002 even as cropland P balances were drawn down, 63–66% of the increase attributable to greater delivery consistent with increased tile drainage and connectivity (Jarvie et al. 2017).

Organic and drained peat soils are distinctive high-risk sources, where wetting/drying, organic-matter degradation, and redox chemistry drive transport (Grenon, B. Singh, et al. 2021). Flooding organic soils increased P release 4–8-fold, and organic-P mineralisation can supply up to 160 kg P ha^{-1} yr^{-1} (Sharpley 1995). Very large filterable P loads were measured from an Organic soil (86.7 kg P ha^{-1}) versus a Podzol (1.7) (McDowell and Monaghan 2015), and degraded drained peatlands were the highest-risk eutrophication source in a German catchment, governed by seasonal, redox-mediated remobilisation (Tiemeyer, Kahle, and Lennartz 2009; Yli-Halla, Lötjönen, et al. 2022). Storm-triggered ochre flocs from tile drains carried 27–141 mg L^{-1} TP with Fe and P co-distributed (Zimmer, Kahle, and Baum 2016).

5.5 Preferential Flow, Macroporosity, and Drainage-Induced Soil Structure

5.5.1 Biopores as the dominant conduits

Rapid delivery of water and surface-applied solutes to drains is governed by a small volume of continuous macropores rather than uniform matrix flow. In a Finnish clay, at 10 cm above the tile movement occurred exclusively through earthworm burrows (106 dyed burrows m^{-2}), the base of the plough layer acted as a hydraulic barrier, and infiltration above the tile lines was about twice that between them and correlated with *Lumbricus terrestris* numbers (Shipitalo, Nuutinen, and Butt 2004). Worm burrows were the only structures at tile depth in a Swiss loam (Stamm et al. 2002), and roughly 50 continuous burrows m^{-2} reached from surface to 90 cm at a German site, their abundance the most important precondition for preferential flow of isoproturon (Zehe and Flühler 2001). Smoke-injection and dye studies confirmed continuous surface-to-drain pathways clustered within a ∼1-m band over the drain, their number increasing as the water table dropped (Nielsen, Petersen, and S. Hansen 2015; Petersen, Nielsen, and S. Hansen 2012). The hydraulically active pore volume is tiny: field analysis indicated an active macroporosity of only $\sim 0.17\%$ of soil volume, yet this fraction produced abrupt drainage concentration increases by factors of 20–400 within hours (Villholth, Jensen, and Fredericia 1998), and macropore area fractions peaked below 0.009, correlating strongly with saturated conductivity (Frey and Rudolph 2011).

5.5.2 Spatial organisation and static structure

Two organising gradients recur: macroporosity and biopore abundance decline sharply with depth, whereas lateral variation relative to the drain is real but weaker. In a Finnish clay, macroporosity fell strongly with depth but differed little with distance, though earthworm burrows were 44% more numerous above a drain and field saturated conductivity varied with drain position (Alakukku et al. 2010; Frey and Rudolph 2011). Reviews report higher infiltration and lower bulk density above drain lines and runoff reductions of 34–55% relative to undrained soils (Montagne, Cornu, Le Forestier, and Cousin 2009). Sustained drainage consistently lowers bulk density and raises air-filled porosity and gaseous diffusivity: Ohio drainage decreased bulk density and increased air-filled pores (HUNDAL, SCHWAB, and TAYLOR 1976), and relative gas-diffusion gains correlated with corn yield (Nakajima and Lal 2014). Consistent with an aeration mechanism, drainage raised soil oxygen, mineralisation, and nitrification (GROFFMAN, BOX, and TODD 1984), and dynamic aeration indicators outperformed static bulk density in explaining productivity across a large Canadian survey (Caron et al. 2025). The effect is context-dependent, however: some comparisons found

lower bulk density under undrained or subirrigated soils (LAL and FAUSEY 1993; B. Baker, Fausey, and K. Islam 2004), drainage reduced surface organic matter and soil organic carbon in some Ohio studies (FAUSEY and LAL 1992; S. Kumar, Nakajima, et al. 2014), and in the most compacted soils aeration control by drainage becomes impossible (SMEDEMA 1993).

5.5.3 Hydraulic evolution, mole drainage, and pedogenesis

Drainage and associated loosening raise infiltration and saturated hydraulic conductivity, and these gains often develop over years. Fields drained > 15 years earlier had saturated conductivity 7.58 cm h^{-1} faster than recently drained fields, with preferential-flow development the likely driver, implying that soil physical and biological properties evolve after installation (Frankl et al. 2023; Jafari-Talukolaee et al. 2016). Where matrix permeability is inadequate, mole drainage and subsoiling create the effective pore network: up to 90% of flow occurred through mole drains on UK heavy clays (HARRIS, HOWSE, and PEPPER 1993), and moled structure persists as open cavities for 3–7 years and thereafter as zones of increased permeability (MULLER 1988; PLAMENAC 1988; Castanheira and Serralheiro 2010). Over years to decades, redirected water fluxes accelerate pedogenesis: subsurface drainage intensified eluviation and clay translocation, removed swelling clays and >one-third of Fe oxides near the drain, and redistributed Mn oxides, with eluviation restricted to the first metre either side of the drain and redox-driven changes extending several metres (Montagne and Cornu 2010; Montagne, Cornu, Le Forestier, Hardy, et al. 2008; Montagne, Cousin, et al. 2013; Mercier et al. 2000). About 70 years of drainage formed an argic horizon in boreal Finland, a change judged probably partly irreversible (Yli-Halla, Mokma, et al. 2009).

5.5.4 Macropore transport of solutes and manure constituents

These structural findings manifest as rapid, connectivity-limited preferential transport. Conservative tracers reach drains within minutes to hours: chloride breakthrough increased by factors of 20–400 within hours (Villholth, Jensen, and Fredericia 1998); bromide appeared within 1 h yet cumulative capture was slow (8% in 48 h, $\sim$ 95% only after months) (Frey, Rudolph, and Conant 2012; Kladivko and Bowling 2023). Liquid-manure constituents reached drains within 5 min, and soluble P moved through artificial macropores with little sorption, identifying macropore flow as the primary rapid-P mechanism (Stamm et al. 2002; Geohring et al. 2001). These behaviours are captured only by models representing two-domain flow or a direct macropore–drain connection: a two-dimensional mobile–immobile model matched bromide dynamics far better than single-porosity formulations (Köhne et al. 2006); adding a macropore domain was essential to match drainage chloride (Villholth and Jensen 1998); an "express fraction" of $\sim$ 2% reproduced immediate tracer breakthrough in RZWQM (Fox, Malone, et al. 2004; Fox, Sabbagh, et al. 2007); and a macropore subroutine in APEX raised DRP-load performance from unsatisfactory to very good, with median annual macropore-flow and macropore-P contributions of 48% and 43% (Ford et al. 2017). Fine-textured soils generated more macropore flow than coarse-textured soils, so soil texture explained connectivity-driven seasonal variation in flow partitioning (Poon, Whalen, and Michaud 2021; Holbak et al. 2021).

5.6 Temporal Dynamics of Phosphorus and Sediment Loss

Phosphorus loss is highly concentrated in time. Most annual loss occurs in the non-growing season, which accounted for 83–97% of runoff, 84–100% of DRP, and 67–98% of TP loss in Ontario (Van Esbroeck et al. 2016; H. Qi et al. 2018). Event dominance is stark: more than 75% of DRP and 66% of TP export occurred in the top 25% of flows, and a single 6-day event supplied 46.6%

of TP, 84.0% of DRP, and 81.0% of suspended solids in one subwatershed (Brendel et al. 2019). Concentration–discharge behaviour differs by pathway: tile TP and DRP increased with discharge (chemodynamic) while surface runoff was more chemostatic, and the tile DRP:TP ratio fell at high flow, indicating a growing PP contribution (Osterholz, Hanrahan, and King 2020). In cold regions, snowmelt and winter thaws dominate exports, predominantly in dissolved form—limiting the value of particulate-focused practices (Baulch et al. 2019; Clement and Steinman 2017). A drought-then-deluge sequence in 2021 produced unprecedented P export (43% of a 12-year DRP load in nine months) (Millar et al. 2025; Mumbi et al. 2025).

Preferential flow reconciles substantial subsurface PP loss with the filtering expected of matrix flow. ^{137}Cs tracing shows drainage particulate matter originates from topsoil (median grain sizes 2–5 μm) translocated within hours (Laubel et al. 1999; Uusitalo, Turtola, Kauppila, et al. 2001), and PP loss is episodic—nearly one-third of annual PP loss occurred during sampled storms (Laubel et al. 1999; Grant et al. 1996). Because of this episodicity, discrete sampling underestimated annual P loss by 26% to > 100% (Grant et al. 1996). At catchment scale, source-fingerprinting attributed 28–55% of sediment yield to field drains, with topsoil supplying 70.9–88.6% of the load and drains contributing over 50% of the annual sediment budget in one catchment (Walling et al. 2002; Chapman et al. 2005).

5.7 In-Field Controls on Phosphorus Loss

Long-term P inputs raise subsurface P losses, generally in the order continuous grass > rotation > continuous corn, with fertilised grass shifting from a P sink to a source (T. Zhang, Tan, Zheng, and Drury 2015). Swine-manure compost produced much greater DRP, PP, and TP than controls (T. Zhang, Tan, Zheng, Welacky, et al. 2015), and about 60% of DRP losses from a no-till cover-crop site followed fall surface fertilisation, highlighting incidental loss (Macrae, Plach, et al. 2023; Mumbi et al. 2025). Across the MANAGE database, generally less than 2% of applied P was lost in drainage (L. Christianson, Harmel, et al. 2016). Tillage involves a trade-off between dissolved and particulate forms: no-till and conservation tillage were associated with higher subsurface DRP, whereas tillage increased PP and sediment (Ni, Y. Yuan, and W. Liu 2020; Macrae, Plach, et al. 2023; Koskiaho et al. 2002). In snowmelt regions, zero tillage and perennial forages can *increase* dissolved P despite reducing erosion (Baulch et al. 2019). Overall, effective P control depends on matching the right practice to site hydrology, connectivity, weather, and legacy P, and on recognising these dissolved–particulate trade-offs (King, Williams, Macrae, et al. 2015; Ni, Y. Yuan, and W. Liu 2020; Baulch et al. 2019).

6 In-Field Nutrient and Cropping Management

Agronomic management of the crop and its nitrogen supply is the principal in-field lever on drainage nitrate, complementary to the drainage-system and edge-of-field practices treated elsewhere. A large plot-, field-, watershed-, and regional-scale literature converges on a consistent picture: drainage nitrate concentration and load are governed jointly by nitrogen (N) input, crop and rotation choice, the amount and seasonal timing of drainage flow, and soil processing of N, so that no single practice is universally decisive but several act in predictable directions (Dinnes et al. 2002). Under conventional maize and maize–soybean systems flow-weighted NO_3^--N routinely exceeds the 10 mg L^{-1} drinking-water standard—more than 70% of site-years in one twelve-state synthesis (W. Liu, Y. Yuan, and Koropeckyj-Cox 2021), and corn, corn–soybean, and soybean site-years averaging 14.0, 13.5, and 12.1 mg L^{-1} across a continental database (Hertzberger et al. 2019). The underlying rate–loss and hydrologic mechanisms are developed in the nitrate-export

section; here the focus is on how rate, source, timing, cover crops, rotation, and tillage alter that loss and at what cost.

6.1 Nitrogen Rate, Source, and Residual Soil Nitrate

Drainage nitrate loss rising with N application rate is the most consistently documented in-field relationship, and it is paired with a yield plateau near the economic optimum. In Iowa production fields, yearly nitrate mass loss was greater for high- than for medium- and low-N treatments, with concentrations of 5–45 mg N L^{-1} (Jaynes, T. Colvin, et al. 2001); four-year losses ranged 22–68 kg N ha^{-1} and rose with rate in both continuous corn and corn–soybean (Helmers, X. Zhou, et al. 2012). Increasing N above the agronomic optimum yields diminishing returns: at 168–252 kg N ha^{-1} in Illinois maize yield differed little while cumulative loads were statistically similar (N. Li et al. 2023), echoing early Iowa work that identified a single 150 kg ha^{-1} application as a favorable compromise (Azevedo et al. 1997); a 10% rate cut lowered yield without significantly reducing loss at one site, underscoring site and hydrologic dependence (Gentry, Green, et al. 2024). Regional syntheses formalize the pattern: maize yield followed a quadratic response peaking near 193 kg N ha^{-1} while drainage N loss increased exponentially with rate, and yield-scaled loss rose sharply once N surplus exceeded $\sim$ 23 kg N ha^{-1}, identifying zero surplus as a practical target (X. Zhao et al. 2016). N rate was the strongest predictor of annual concentration ($R^2 = 0.59$) with no yield gain above 150–224 kg ha^{-1} (Hertzberger et al. 2019), the most cost-effective high-rate scenarios also produced the highest losses (W. Liu, Y. Yuan, and Koropeckyj-Cox 2021), and across syntheses each unit rise in rate produced $\sim$ 0.12 kg NO_3^--N loss, rate explaining 27% of variability (Chatterjee 2020; L. Christianson and Harmel 2015; Ritter 2015).

Residual soil nitrate (RSN) links rate to leaching risk and creates legacies. RSN was minimally affected below the economic optimum N rate (EONR) and increased sharply only above it (Bandura et al. 2025); continuous corn accumulated up to 225 kg ha^{-1} RSN in dry years, flushed to drainage in subsequent wet years (Randall and Mulla 2001), and a very high fall rate continued to affect nitrate and residual N for more than four years (Jaynes 2015). Encouragingly, corn yields rose over three decades at nearly constant rate while drainage nitrate stayed flat, indicating improved use efficiency but no loss reduction from yield gains alone (Clifford et al. 2023); the leaching breakpoint lies above the agronomic optimum, with a larger buffer in rotations (Pasley et al. 2021).

Nitrogen source matters, with manure generally increasing loss relative to inorganic fertilizer at comparable rates. Liquid swine manure produced greater concentrations and losses than UAN with no yield difference (Bakhsh, Kanwar, and Karlen 2005), liquid-hog manure raised soil and drainage nitrate above inorganic fertilizer with equivalent yields (Elmi, Madani, et al. 2005), and annually repeated fall manure produced the highest losses, timing of manure addition controlling loss more than rotation (Hernandez-Ramirez et al. 2011). Applying manure before both crops rather than only corn raised loads (B. Dougherty et al. 2020), and fall manure leached more than spring manure (Bakhsh, Kanwar, Pederson, et al. 2007; Rekha et al. 2011). Organic sources generally boosted yield but increased losses (W. Liu, Y. Yuan, and Koropeckyj-Cox 2021), though one meta-analysis found no significantly greater dissolved N load (L. Christianson and Harmel 2015).

6.2 Nitrogen Timing, Placement, and Enhanced-Efficiency Products

Shifting N from fall to spring or in-season application generally lowers drainage nitrate, but effects are smaller and less consistent than rate or cropping-system changes. In Minnesota rotations fall N without nitrapyrin produced the highest losses, with reductions relative to fall of 13–18% (spring, split, fall+nitrapyrin) (Randall, Vetsch, and Huffman 2003; Randall and Vetsch 2005); fall

N produced greater concentration and load than spring N in Illinois (Gentry, Green, et al. 2024), and spring or side-dress N cut concentrations while maintaining yield (Pittelkow et al. 2017; Vetsch, Randall, and Fernández 2019). Minnesota work found spring application raised use efficiency > 20% and reduced tile losses ∼ 36% relative to fall (Dinnes et al. 2002), though the fall-versus-spring effect was often smaller than weather, drainage volume, and rate (W. Liu, Y. Yuan, and Koropeckyj-Cox 2021), and out-of-season and pre-plant timings had higher concentrations than at-plant and side-dress timings (Hertzberger et al. 2019).

Split, side-dress, and placement effects are mixed. Split fertilization reduced export only at lower rates (W. Liu, Y. Yuan, and Koropeckyj-Cox 2021) and was not significantly different from non-split in a continental synthesis (Hertzberger et al. 2019); side-dress at V2–V12 gave no consistent difference (Jaynes 2013), and midseason splits sometimes raised concentration and RSN (Jaynes and I. Colvin 2006; Bandura et al. 2025). Split-applied urea modestly improved water quality (Waring, Sawyer, et al. 2022), and remote-sensing variable-rate side-dress averaged a 40% reduction over 15 years, mainly via 33% lower input and avoidance of peak flow, while maintaining yield (G. Wilson, Mulla, Galzki, et al. 2020). Placement was inconsistent: injection and incorporation sometimes had higher concentrations than surface application (Hertzberger et al. 2019), and a localized-compaction applicator gave 16% lower concentrations than a spoke injector (Bakhsh, Kanwar, and J. Baker 2010). Among enhanced-efficiency products, nitrapyrin generally did not reduce nitrate or improve yield alone (Vetsch, Randall, and Fernández 2019; Pittelkow et al. 2017) but reduced fall-N losses 10–18% in Minnesota rotations (Randall, Vetsch, and Huffman 2003; Randall and Vetsch 2005); polymer-coated products were more robust, with ESN lowering combined N_2O, NH_3, and NO_3^- losses 47–49% below urea while maintaining yield, the largest benefit in wet periods (Menegaz et al. 2025; P. Nash, Nelson, Motavalli, and S. Anderson 2015; P. Nash, Nelson, and Motavalli 2015). Reduced-rate summer fertigation cut concentration and load ∼ 16% without reducing silage yield (J. Gamble et al. 2018), and fall anhydrous ammonia with nitrapyrin matched spring application environmentally (H. Kaur, Nelson, G. Singh, and G. Kaur 2024; H. Kaur and Nelson 2024).

6.3 Cover Crops: Reductions, Mechanisms, and Yield

Cover crops—particularly fall-planted winter cereal rye—are among the most consistent and largest in-field reducers of drainage nitrate. The clearest, most repeatable signal is reduced concentration. In a long-term Iowa study rye cut concentrations 48% over five years and oat 26%, though neither significantly reduced cumulative loads because annual drainage varied (Kaspar et al. 2012); rye reduced concentrations > 30% even at low biomass (Waring, Lagzdins, et al. 2020), and reviews report reductions of 13–94% (Ward et al. 2018). Where rye is well established, large load reductions follow: split N plus rye reduced flow-weighted concentration and load 35% and 37% without yield penalty (Preza-Fontes, Pittelkow, et al. 2021); rye added to spring and fall systems cut load 40% and 47%, reaching 61–64% in a wet year (Ruffatti et al. 2019); a no-till rye system lowered losses 59–67% (O'Brien et al. 2022); rye contributed a 43% decrease in cumulative four-year load (Gentry, Green, et al. 2024); and Indiana watershed monitoring found field-scale reductions of 58–99%, pooling to 65% (Vincent et al. 2025). Multi-year and northern-margin studies show meaningful but often statistically marginal cumulative effects: rye reduced cumulative drainage 12.1% and N loss 20.4% over six Iowa years, with significant reductions only in favorable years (Martinez-Feria et al. 2016); 11% discharge and 13% loss reductions in Minnesota (Strock, P. Porter, and Russelle 2004); and in a manured silage system rye lowered concentration from 53 to 39 mg L^{-1} but loads remained high, warranting additional practices (Krueger et al. 2013).

The mechanism is assimilation of soil mineral N into biomass during the fall–spring fallow, shifting nitrate out of leaching-vulnerable depths. Cereal rye and radish accumulated 130–250 kg N ha^{-1}

and moved NO_3-N from environmental (20–80 cm) to agronomic depths, high-C/N overwintering rye best protecting N in wet springs (Lacey and Armstrong 2015); living cover consistently lowered soil and vadose-water nitrate (Bosompemaa et al. 2021; Strock, P. Porter, and Russelle 2004; Z. Qi, Helmers, et al. 2011). Effectiveness is therefore strongly biomass- and weather-dependent: biomass and reductions were weak under poor conditions (Preza-Fontes, Pittelkow, et al. 2021; O'Brien et al. 2022), cereal rye biomass varied 1.4–5.9 Mg ha^{-1} (highest under late termination) (Gentry, C. Mitchell, et al. 2025; Birru et al. 2023), and success depended on early establishment, achieved in only one of four years after corn in one Minnesota study (Strock, P. Porter, and Russelle 2004; Bawa et al. 2023). Regional modeling explained $> 95\%$ of site-to-site variation, rye being more effective in warmer, less-wet southern locations (Malone, Jaynes, Kaspar, et al. 2014). Effects on drainage *volume* are modest and often non-significant, so load reductions are driven mainly by concentration: rye cut May–July drainage 21% via enhanced spring evapotranspiration but annual drainage only 9% (non-significant) (Z. Qi and Helmers 2010; Z. Qi, Helmers, et al. 2011), and in the 2012 drought rye did not deplete soil water at the cash crop's expense (Daigh, Helmers, et al. 2014).

Rye cover cropping generally imposes no consistent yield penalty, especially for soybean (Strock, P. Porter, and Russelle 2004; Waring, Lagzdins, et al. 2020; Z. Qi, Helmers, et al. 2011), with projected-climate differences of only ~ 0.1 Mg ha^{-1} and no continuous-corn penalty in simulation (Malone, Garbrecht, et al. 2020; Birru et al. 2023). Documented exceptions concentrate in maize, drought years, and silage—a 6% average maize reduction (34% in 2012) (Martinez-Feria et al. 2016) and 16% lower corn silage after manured rye (Krueger et al. 2013). Management modulates outcomes: earlier fall sowing is consistently more effective (mid-September sowing reduced losses 45% versus October dates) (Feyereisen, B. Wilson, et al. 2006), and delayed spring termination raises biomass with limited yield cost (Birru et al. 2023; Chatterjee et al. 2025). In harvested-rye systems, fertilizing the rye boosts biomass and energy with only a small N-loss cost (Malone, Obrycki, et al. 2018; Malone, Radke, et al. 2023). Synthesizing across studies, well-established rye clusters around 40–60% N-loss reduction with 25–50% concentration reduction (Malone, Obrycki, et al. 2018; Chatterjee et al. 2025; Malone, Jaynes, Kaspar, et al. 2014; Malone, Radke, et al. 2023); Iowa climate-change projections had rye reduce annual loads from 47.5 to 18.9 kg N ha^{-1} (60%) under baseline and 63.6 to 17.0 kg N ha^{-1} (73%) under future climate, meeting load targets for all six climate models (Malone, Garbrecht, et al. 2020), and watershed models estimate 15–34% reductions in tile nitrate yield (David et al. 2015; Makhtoumi, Aragon, and Lark 2025; R. Gupta, Bhattarai, Dokoohaki, et al. 2023).

6.4 Crop Rotation, Diversification, and Perennial Systems

Cropping system exerts strong control on drainage nitrate, generally exceeding tillage and often rivaling N rate. Continuous corn consistently lost more nitrate than corn–soybean or perennial systems, leading authors to judge it environmentally unsustainable (Weed and Kanwar 1996; Kanwar, T. Colvin, and Karlen 1997); rotations produced greater yield and lower yield-scaled loss and a larger buffer above the agronomic optimum (X. Zhao et al. 2016; Pasley et al. 2021). Combining a shift to corn–soybean with reduced rate and a cover crop lowered concentration from 28 to 8 mg L^{-1} over 14 years (Ritter 2015), and more diverse rotations delivered large reductions with maintained yield—a three-year rotation averaged 4.2 versus 8.0 mg L^{-1} for a control, exceeding 10 mg L^{-1} in only 1% of samples (Gentry, C. Mitchell, et al. 2025; Ahmed et al. 2007).

Perennial and prairie systems are the most effective in-field option. Prairie concentrations were ≤ 1.4 mg L^{-1}, 2–195 times below row crops (Daigh, X. Zhou, et al. 2015); restored prairie grass reduced losses to near rainfall levels (Hernandez-Ramirez et al. 2011); and alfalfa and CRP plots

had 0.3–4 versus 14–40 mg L^{-1} for row crops, with four-year losses of $\sim$ 200 kg ha^{-1} for corn versus 4–7 for perennials (Randall and Mulla 2001; Waring, Pederson, et al. 2024). The benefit can be short-lived on conversion back to row crops as RSN rises (Huggins, Randall, and Russelle 2001). Converting corn–soybean area to alfalfa or wheatgrass reduced watershed losses 78–81% (G. Wilson, Mulla, Jordan, et al. 2023), and bioenergy perennials also reduce loss (Ferin et al. 2023; Woodbury et al. 2018; Guo et al. 2018). Diversified rotations, cover crops, and perennials are recognized as complementary reducers (Dinnes et al. 2002).

6.5 Tillage and Residue Management

Tillage exerts weak and inconsistent control over drainage nitrate relative to precipitation, cropping system, and N rate. Over 11 Minnesota years drain flow was higher under no-till but nitrate losses only slightly higher under conventional tillage, and moldboard-plow and no-till losses were similar (43 and 41 kg ha^{-1}), cropping system mattering far more (RANDALL and IRAGAVARAPU 1995; Randall and Mulla 2001). In Iowa no-till and ridge-till usually had the lowest losses but rotation had greater influence (Weed and Kanwar 1996; Kanwar, T. Colvin, and Karlen 1997), regional syntheses found no significant tillage difference in N load (X. Zhao et al. 2016), and conservation tillage had the highest concentration (16.1 mg L^{-1}) (Hertzberger et al. 2019; Z. Wang, Z. Qi, et al. 2016). Where reduced tillage builds soil health its nitrate effect remains equivocal: no-till lowered loads only in some water years and reduced yields (Masri, Asher, and Piwarski 2024), though it raised soil organic carbon and lowered EONR (Cassman et al. 2025; Waring, Pederson, et al. 2024). Residue and stover management had limited effect on nitrate: 50% stover harvest reduced simulated loss only 1.1 kg ha^{-1} (Malone, Herbstritt, et al. 2019), residue removal had little impact on concentration or yield (B. Dougherty et al. 2020; Guo et al. 2018), and continuous corn with residue removal plus rye achieved losses similar to other row crops (Daigh, X. Zhou, et al. 2015; Farahani, Asoodar, and Moghadam 2020).

6.6 Soil Testing, Use Efficiency, Regional Aggregation, and Loss-Pathway Trade-offs

Watershed-scale soil-test-based programs confirm in-field mechanisms at landscape scale. Adopting the late-spring nitrate test (LSNT) across a 400-ha Iowa subbasin reduced concentration by at least 30% via reduced fertilizer use while matching yields (Jaynes, Dinnes, et al. 2004; Karlen et al. 2005; Malone, Jaynes, L. Ma, et al. 2010), and soil-test programs generally reduced RSN and loss versus yield-goal management (Dinnes et al. 2002; Ritter 2015). Alternative farming practices reduced drainage 41% and concentration from 17.2 to 8.2 mg L^{-1}, cutting losses 59–62% (Oquist, Strock, and Mulla 2007), and because 85% of a Minnesota watershed's load came from tile drainage, substantial perennialization was needed for a 30–50% reduction (G. Wilson, Mulla, Jordan, et al. 2023; Makhtoumi, Aragon, and Lark 2025). Use efficiency framing is increasingly emphasized: partial factor productivity was highest for side-dress and lowest for very high fall N (Jaynes 2015), and EONR is proposed as a sustainability standard (Bandura et al. 2025); adoption of cover crops, controlled drainage, and bioreactors remains constrained by financial and social barriers (David et al. 2015).

Scaling site results to the drained landscape indicates regionally significant potential. For the five-state Corn Belt, winter rye was estimated to reduce nitrate loss to the Mississippi by $\sim$ 20%, with adoption on 34–81% of agricultural land and removal costs of US\$2.24–3.25 kg^{-1} N (Kladivko, Kaspar, et al. 2014); a statewide Illinois analysis found rye reduced tile nitrate loss $\sim$ 30% (R. Gupta, Bhattarai, Kalita, et al. 2023), and a five-state energy-cover-crop assessment estimated a

216 million kg yr^{-1} reduction (27% of the assumed Gulf load), with nine hotspot clusters on 19% of the area contributing 48% of the reduction (Malone, Radke, et al. 2023).

Finally, mitigation practices redistribute N among loss pathways—leaching, N_2O, and NH_3—so single-pathway evaluation can mislead. In-season N plus rye cut leaching 37% but raised annual N_2O 26%, leaving total N loss and environmental-damage cost statistically unchanged (Preza-Fontes, L. Christianson, and Pittelkow 2023). Yet split N plus rye had the lowest dissolved N_2O and nitrate loads, and no consistent nitrate–N_2O tradeoff emerged across systems, showing stacked practices need not trade one pathway for another (Oliveira, Foltz, et al. 2025; O'Brien et al. 2022); ESN lowered N_2O, NH_3, and NO_3^- simultaneously (Menegaz et al. 2025). N_2O rose nonlinearly with rate above a threshold near 120 kg N ha^{-1}, and each unit of fertilizer-N produced 0.01 kg N_2O-N alongside 0.12 kg NO_3^--N loss, subsurface-drained fields having the highest mean N_2O emission (Z. Wang, Z. Qi, et al. 2016; Chatterjee 2020; Fontes et al. 2019). The consistent conclusion across scales is that combining two or more practices—rate optimization, in-season timing, cover crops or diversified rotation, and drainage management—reduces nitrate loss most cost-effectively, since in-field management alone is often insufficient (Dinnes et al. 2002; S. Singh, Hwang, et al. 2023).

7 Drainage Design and Water-Table Management

Drainage design and active water-table management are the primary levers through which practitioners set both the volume of subsurface outflow and the loads it carries. Because solute export in drained cropland is governed first by hydrology, the geometric choices of spacing and depth and the operational choice of when to hold or release water translate directly into edge-of-field flow and nutrient loss. This section synthesises what the evidence has established about those controls: how spacing and depth trade flow against loss, how controlled drainage and drainage water management reduce loads in proportion to the flow they retain, how capture-and-reuse systems recover both water and quality benefits, and how enhanced configurations and controlled drainage perform under projected climates.

7.1 Drain Spacing and Depth as Controls on Flow and Loss

Drain spacing and depth are well-established design controls on both drainage volume and nitrate loss. More closely spaced drains produce greater drainage and greater nitrate-N loss than wider spacings, whereas nitrate *concentrations* generally do not vary consistently with spacing—load differences are driven by flow rather than by concentration (Kladivko and Bowling 2021; Kladivko, Frankenberger, et al. 2004). Calibrated models consistently show that shallower and wider-spaced drains reduce tile flow and nitrate loss at modest yield cost: reducing drain depth from 1.5 to 0.9 m lowered simulated nitrate loss from 34.0 to 29.4 kg ha^{-1}, and widening spacing from 27 to 60 m lowered it from 33.8 to 28.4 kg ha^{-1} (Moriasi et al. 2013; Nangia et al. 2010; D. Davis et al. 2000; Hwang et al. 2024). The design tension is that reducing subsurface loss shifts water to surface runoff: for phosphorus, increasing spacing reduced subsurface volumes and tile TP loss (more strongly on finer soils) while increasing surface runoff and its associated TP load (Morrison, Madramootoo, and Chikhaoui 2013).

These relationships motivate an optimum-intensity principle: use the minimum subsurface intensity required to relieve excess-water stress, fit the system to the soil, and apply water-table control where maximum intensity is unnecessary (Breve et al. 1998). Deeper drains often raise profit but substantially increase nitrate losses, and simulation showed that widening spacing to 100 m cut total nitrate-N loss by 20% (37 to 29 kg ha^{-1}) at small profit cost (El-Sadek, Feyen, W. Skaggs, et al. 2002; Breve et al. 1998); deep drains are frequently unnecessary for salinity control

(Ritzema and Schultz 2011). Shallow drainage reproduces much of the flow- and load-reduction benefit, reducing Iowa subsurface volume by 46–58% and nitrate by 29–49% generally without reducing yield (Helmers, R. Christianson, et al. 2012; Schott et al. 2017). A multi-state analysis found shallow drains gave quicker water-table drawdown, reduced yield variability, and 3.4 cm less total flow than deep drains, though with more surface runoff and a lower benefit–cost ratio (Ghane and Askar 2021), and field comparisons found shallow systems had the lowest nitrate concentration and loading among configurations, sometimes at a yield penalty whereas controlled drainage trended toward higher yields (E. Smith, Vosman, et al. 2019; E. Smith and Kellman 2011b).

7.2 Controlled Drainage and Drainage Water Management

Controlled drainage (CD)—raising an outlet control structure to hold the water table between drainage events—is among the best-established conservation-drainage practices. A synthesis found CD reduced drainage outflow by 17% to more than 80% depending on soil, drainage intensity, control strategy, and location, with nitrogen-loss reduction often about the same percentage as the volume reduction (R. Skaggs, Youssef, Gilliam, et al. 2010). A global meta-analysis of 84 publications and 1788 paired site-years confirmed that CD and controlled drainage with subirrigation (CDSI) reduce subsurface discharge and nutrient losses while modestly increasing yield; CDSI reduced nitrate and TN more, CD reduced TP more, and CDSI (but not CD) significantly increased surface runoff and surface nitrate loss (R. Wu et al. 2025). Simulations across 48 Midwest locations found CD reduced drainage and N losses everywhere, with efficacy strongly climate-dependent (Youssef et al. 2018).

Field studies quantify these effects concretely. Quebec CDSI cut drainage volume 24%, nitrate concentration 25%, and nitrate loss 43% (Ng et al. 2000); a long-term Canadian study found CD reduced total nitrate loss by $\sim$ 58% and increased yield 3% without elevated greenhouse-gas emissions (Madramootoo, Essien, and Ekwuknife 2024); and nine years of North Carolina CD reduced outflow 33% and nitrate export 30% while raising yields 10–11% (Poole et al. 2018). Growing-season CD reduced water flux $\sim$ 60%, nitrate 51%, ammonium 58%, DRP and TP 66%, and *E. coli* 76%, with watershed-scale deployment cutting stream fluxes of water, N, and P (Sunohara, Gottschall, Craiovan, et al. 2016; Sunohara, Gottschall, Wilkes, et al. 2015). In Illinois, managed systems always exported less nitrate than free drainage, with flow reductions of 35–96% and load reductions of 37–79% (averaging 61%) (Cooke and S. Verma 2012).

The mechanism is consistently one of flow–load coupling: syntheses report drainage water management reducing tile flow by $\sim$ 46% and nitrate loads by $\sim$ 45–48% *mainly by reducing discharge rather than by enhancing denitrification*, which plays only a secondary role (Ross et al. 2016; Ward et al. 2018). Two trade-offs qualify the benefit. First, because reduction operates through volume, drainage management often fails to lower nitrate *concentrations* even where it lowers loads (Grenon, Madramootoo, et al. 2023; Wesström, Joel, and Messing 2014). Second, CD can increase surface DRP and TP that partly offset subsurface benefits (King, Hanrahan, et al. 2022; R. Wu et al. 2025). Timing is decisive: DRAINMOD analysis identified early March as the most effective control window (reductions up to 100%) whereas April starts resembled free drainage (Kesicka, Kozlowski, and Stasik 2023). Greenhouse-gas responses are mixed but generally not adverse (Madramootoo, Essien, and Ekwuknife 2024; E. Smith, Vosman, et al. 2019).

7.3 Drainage Water Recycling and Subirrigation

Drainage water recycling (DWR)—capturing and storing drainage for supplemental irrigation—benefits both production and downstream water quality. Yield gains reach 61% for corn and

31% for soybean, with Canadian N-loss reductions of 40–70% and P-loss reductions of 12–36%, and modelled downstream N-delivery reductions of 24–37% (Hay et al. 2021). A North Carolina reservoir reduced TN load 47%, TP 30%, and sediment 87%, with hydraulic retention time the key control, although the reservoir was a net source of particulate P (Moursi, Youssef, Poole, et al. 2023; Moursi, Youssef, and Chescheir 2022). Resilience is a defining advantage: multi-location synthesis of 53 site-years showed DWR is more resilient than free drainage, protecting yield during both drought and excess precipitation and reducing yield variability by 22% (Willison et al. 2021). Free drainage and drainage water recycling raised corn yield 20% and 26% respectively over a nondrained control (recycling gains reaching 48% in individual years) by avoiding drought losses (H. Kaur, Nelson, and G. Singh 2021), and North Carolina recycling raised corn and soybean by 8% and 17% with subirrigation raising the water table 15 cm (Moursi, Youssef, and Poole 2024). Adoption economics indicate crop insurance is not a major barrier (Sellars et al. 2022), and in arid systems integrated on-farm drainage management reused all but 0.7% of applied water (Ayars and Soppe 2014; Kosari et al. 2024).

Subirrigation—delivering water upward through the drains—similarly lowers losses and stabilises supply. It reduced soil-profile nitrate by up to 50% and drainage nitrate concentrations by 55–73%, with water-quality benefits accruing even when outflow occasionally exceeded free drainage, implicating enhanced denitrification (Elmi, Madramootoo, Egeh, and Hamel 2004; Elmi, Madramootoo, Egeh, Dodds, et al. 2002; Noory and Liaghat 2009). CDSI raised groundwater levels, supplied 170–410 mm of infiltration, and increased Dutch yields by up to 64% (Wit, Ritsema, et al. 2022; Wit, Huijgevoort, et al. 2024; FAUSEY, BROWN, et al. 1995; Almen et al. 2021; Madramootoo, Johnston, et al. 2007). Its performance, however, is strongly site-dependent, governed by lateral hydraulic conductivity: across three Ohio sites subirrigation improved relative yield to 87–93% and 80–91% at two sites but had almost no effect at Defiance owing to very low conductivity (Gunn, Baule, et al. 2018; Gunn, Allred, et al. 2019). Where hydraulics are favourable, CD and subirrigation kept soil water optimal in both wet and dry years (Niaghi et al. 2019), and in claypan soybean systems subirrigation improved yield, reduced interannual variability, and allowed earlier planting (Nelson, Smoot, and Meinhardt 2011); combining drainage water management with foliar pyraclostrobin produced synergistic soybean gains up to 1070 kg ha^{-1} (Nelson and Meinhardt 2011).

7.4 Enhanced Configurations and Persistence under Climate Change

Beyond conventional single-level layouts, engineered configurations extend the same design principles. Multi-level (bilevel) systems cut the time the water table exceeded 0.5, 0.75, and 1.0 m by 84%, 67%, and 41% while producing much lower shallow-drain salinity (Hornbuckle, Christen, and Faulkner 2007); analytical theory shows that adding shallow drains between deep drains raises the drainage rate and allows wider deep-drain spacing (Hornbuckle, Christen, and Faulkner 2012), with managed deep drains occupying an intermediate position (Ayars, Christen, and Hornbuckle 2006; Christen and Skehan 2001). Engineered enhancements alter flow further: a permeable filter increased discharge (about 1.9 times conventional) and accelerated drawdown while reducing runoff (Tao, S. Wang, D. Xu, and Qu 2016; Tao, S. Wang, D. Xu, H. Yuan, et al. 2017); combined open-ditch-plus-subsurface designs improved waterlogging removal (Ren, S. Wang, P. Yang, Tao, and H. Chen 2021; Tao, J. Liu, et al. 2025); siphon-type drainage raised flow rates 27–90% (Ren, S. Wang, P. Yang, and Tao 2023); artificial macropore tubes generated flow where none occurred and desalinated the profile (W. Chen et al. 2021); and supplementary drainage of a clay field raised mean annual discharge 2.6-fold and cut topsoil runoff by 75% (Häggblom et al. 2019).

A robust and management-relevant finding is that the benefits of controlled drainage persist,

and in some seasons strengthen, under projected climates. In Ohio, conventional discharge was projected to decline while CD annual reductions remained near 4–6% and summer growing-season reductions increased, indicating greater warm-season value (Pease, Fausey, et al. 2017). Michigan simulations projected free-drainage discharge rising 20% by mid-century while CD continued to reduce discharge by 63–72%, with future and historical performance statistically indistinguishable (Bin Shokrana, Ghane, Abdalaal, et al. 2023). Finnish scenarios projected CD effects remaining similar to the historical interval (Salla, Salo, and Koivusalo 2022), and Canadian projections of warmer, wetter conditions found CD reducing subsurface drainage more strongly than free drainage, though rising temperature drove dry-stress yield reductions that motivate the capture, storage, and reuse of drainage water (Ndulue and Ranjan 2024; Gunn, Baule, et al. 2018). Chinese and Polish studies likewise found that appropriate outlet-blocking strategies could reduce future runoff and maintain groundwater at present levels (Awad, Luo, and Zou 2021; Sojka et al. 2020). Together these results indicate that the flow–load coupling underpinning water-table management is durable: because controlled systems act on discharge, their proportional benefits carry forward into a wetter, warmer future.

8 Edge-of-Field Treatment, Reactive Media, and Watershed-Scale Integration

Once nutrients enter tile drains, in-field management can no longer reach them, so a family of edge-of-field practices intercepts the load before it reaches surface waters. Two decades of microcosm and column studies, pilot- and full-scale installations, and global syntheses have established constructed wetlands, denitrifying woodchip bioreactors, reactive phosphorus filters, saturated buffers, and in-channel structures among the best-characterised measures for artificially drained cropland. Their performance is governed by a small, recurring set of controls—hydraulic residence time (HRT), temperature, flow, loading, pH, and cumulative loading—and their reductions must be interpreted against the fraction of drainage actually treated and against downstream, watershed-scale targets. Controlled drainage and drainage water recycling, treated in the design section, complement these practices.

8.1 Constructed wetlands

Constructed wetlands treating drainage are extensively documented nitrate sinks in which denitrification dominates. Across French watersheds, nitrate removal averaged $\sim 42\%$, of which denitrification supplied 93%, and devoting $\sim 1\%$ of the contributing area can achieve 50% removal, with hydraulic residence time the priority design control (Tournebize, Chaumont, and Mander 2017). Field systems span a coherent range: Iowa wetlands reduced incoming nitrate 49% (Drake et al. 2018), a Minnesota wetland removed 67% over four years (Gordon et al. 2021), and Illinois wetlands retained 36–37% of N, implying ~ 450 ha to cut watershed N loading 46% (Kovacic et al. 2006). Broader syntheses place average total-N removal near 39% (H. Wang and Ishii 2025), with roughly 40–50% N removal generally achievable at wetland-to-catchment area ratios of 1–4% and only minor gains beyond (Mendes 2021). Because performance is a strong function of hydraulic loading, residence time, temperature, and inlet concentration, no single-number design is adequate, and higher influent loads raise removal *rate* but lower percentage efficiency (Kadlec 2012; Steidl, Kalettka, and Bauwe 2019; Mendes 2021). Nitrate removal is reliable, but phosphorus performance is variable and wetlands sometimes export more TP than they receive (Tanner, Nguyen, and Sukias 2005; Tanner and Sukias 2011; Messer et al. 2021), with plant harvest an important P route in

some systems (Lenhart et al. 2016). Paired surface-flow wetland plus woodchip systems act as consistent nutrient sinks (TN removal 8–51%, TP 0–80%), the woodchip component contributing more removal than the sedimentation basin while occupying a smaller footprint than a standalone wetland (Pugliese, Skovgaard, et al. 2020).

8.2 Woodchip bioreactors: performance, controls, and mechanisms

8.2.1 Field-scale nitrate removal

Denitrifying woodchip bioreactors reduce nitrate dependably across climates and designs, with a magnitude that spans a wide but well-explained range. Consolidating full-size systems worldwide, L. Christianson, Cooke, et al. (2021) report a mean nitrate-N load reduction of $40 \pm 26\%$ (median 46%) and removal rates of 7.2 ± 9.6 g N m^{-3} d^{-1}, removing an estimated 20–40% of annual nitrate-N loss in the Midwest. The largest compilation—36 site-years at ten full-size Illinois bioreactors—found a median annual removal of 49 kg N per bioreactor, a median edge-of-field efficiency of 17%, and removal in *every* year (L. Christianson, Hay, et al. 2026). Controlled and cold-climate studies fill in the range: an Illinois field experiment averaged 63% (20% at 2-h to 98% at 8-h HRT; 11.6 g m^{-3} d^{-1}) (Bell et al. 2015); a Quebec system cut flow-weighted nitrate-N load 99% and total-N 72–73% (Husk, B. Anderson, et al. 2017); four South Dakota bioreactors reduced loads 39–89% at 5.2–26.5 g N m^{-3} d^{-1} (Thapa et al. 2023); and a New Zealand system rose from 31% (2.8-d HRT) to 93% (6.4-d) (Rivas et al. 2020). Woodchip *denitrification walls* are equally well supported: an Iowa wall reduced annual nitrate loss from 54.5 to 24.5 kg ha^{-1} over nine years with no significant trend (Moorman, Parkin, et al. 2010), a wall cut flow-weighted nitrate and mass loss $\geq 55\%$ with no yield penalty and no land taken out of production (Jaynes, Kaspar, et al. 2008), and a rye cover crop plus wall reduced N leaching 58–59% (Rogovska et al. 2023). Not all systems meet targets: underloaded or low-input bioreactors removed only $\sim 9.5\%$ (Bock, Coleman, and Easton 2018) or 6–28% of edge-of-field nitrate (Maxwell, Cooke, and L. Christianson 2022).

8.2.2 Controls: hydraulics, temperature, and loading

The dominant, repeatedly quantified control is HRT, which trades off against the fraction of flow captured: efficiency rises with HRT (Bell et al. 2015; Rivas et al. 2020), while in the ten-bioreactor dataset annual flow-weighted HRT correlated with flow captured ($\rho = -0.67$) and treated-water removal efficiency ($\rho = 0.60$) (L. Christianson, Hay, et al. 2026). Column work shows removal rate per unit wood rising with flow while efficiency falls (Greenan, Moorman, Parkin, et al. 2009), and higher influent loads raise the removal *rate* but lower percentage efficiency (Rivas et al. 2020; L. Christianson, Hay, et al. 2026; F. Zhang et al. 2021). Temperature exerts a strong, consistent effect: exponential functions of water temperature and HRT explained 73% of the variance in load reduction in Illinois (Bell et al. 2015); South Dakota removal fell $\sim 30\%$ below 12 °C (Thapa et al. 2023); columns dropped from 96% at 21.5 °C to 29% at 10 °C (Soupir et al. 2018); and Michaelis–Menten V_{max} fell sharply from 22 to 5 °C (Kouanda and Hua 2021; F. Zhang et al. 2021). Kinetically, removal is zero-order when nitrate is non-limiting and shifts to first-order once nitrate becomes limiting (Hua et al. 2016; Kouanda and Hua 2021).

8.2.3 Mechanisms, carbon, and microbial ecology

Denitrification is firmly established as the dominant pathway, accounting for $> 98\%$ of transformed nitrate in microcosms, with immobilisation and DNRA minor (Greenan, Moorman, Kaspar, et al. 2006; Greenan, Moorman, Parkin, et al. 2009). Woodchip carbon sustains this capacity for years:

wood raised denitrification potential 31- to 4000-fold over subsoil, supported denitrifier populations $> 10^8$ g^{-1}, and gave in situ nitrate half-lives of 0.64–0.81 d (Moorman, Parkin, et al. 2010); DNRA and anammox occur secondarily (Hartfiel, Schaefer, et al. 2022; Mendes 2021). Cellulose is the most rapidly consumed compound, and wood decomposition is depth-dependent (up to 75% loss near the inlet/top, $< 13\%$ at depth), governing longevity (Hartfiel, Schaefer, et al. 2022; Moorman, Parkin, et al. 2010; Ghane, Feyereisen, et al. 2018). Alternative substrates can raise performance: corncob (CC75) removed more N than woodchips (61.2% vs 37.9%) at lowest cost (Law, Slade, et al. 2023), corn-stover blends enhanced removal (Oladeji et al. 2023), and agricultural residues can outperform woodchips for removal though woodchips give better hydraulics (Greenan, Moorman, Kaspar, et al. 2006; Kohn et al. 2023; F. Zhang et al. 2021). Communities are shaped by substrate, depth, season, and temperature (Hartfiel, Schaefer, et al. 2022), with cold-active denitrifiers (*Cellulomonas*, *Pseudomonas*) and nitrate-reducing fungi identified as key taxa (E. Anderson et al. 2020; Jang et al. 2019; Aldossari and Ishii 2021); bioaugmentation shows conditional promise but is limited by washout (Jeglot et al. 2022; H. Wang and Ishii 2025), and denitrification function is generally robust to water-level disturbance (Hathaway, Bartolerio, et al. 2017).

8.2.4 Design, sizing, and enhancement

Sizing balances flow treatment against efficiency: using 10–20% of peak flow at 6–8-h design retention treats the majority of annual volume, and longer, narrower beds outperform short, wide ones (L. Christianson, R. Christianson, Helmers, et al. 2013). Load-duration analysis showed beds sized to a 0.5-d HRT can remove 44–52% of base-flow nitrate and 20–30% of total annual loss while occupying $< 0.27\%$ of watershed area (Moorman, Tomer, et al. 2015). Hydraulically, high-aspect-ratio designs approach plug flow, whereas wide designs suffer dispersion and short-circuiting (Oliveira, H. Dougherty, et al. 2023; H. Dougherty et al. 2020); expanding cross-sectional area improves flow capture (Maxwell, Cooke, and L. Christianson 2022). Enhancement strategies extend or intensify treatment: pumping supplies additional nitrate-laden water and lengthens the season (Hartfiel, Craig, and Soupir 2023; Díaz-García et al. 2025), electrical stimulation raises removal at a cost (Law, Soupir, et al. 2018), and heated tourmaline reduces clogging while raising removal (C. Ma et al. 2021). A catchment-scale three-bed system removed 713 kg nitrate-N (12.5% of catchment load) but required a sediment-exclusion system during high flow (Feyereisen, Ghane, et al. 2023).

8.3 Pollution swapping and greenhouse-gas by-products of bioreactors

Non-ideal conditions can generate N_2O, CH_4, sulfate reduction, and dissolved-organic-carbon (DOC) losses, but reported risks are generally small (Hartfiel, Schaefer, et al. 2022). Nitrous-oxide emissions are consistently a tiny fraction of nitrate removed: 0.003–0.028% of N denitrified in columns (Greenan, Moorman, Parkin, et al. 2009); no significant difference from controls at an Iowa wall (indirect emission equivalent to tile drainage without the bioreactor) (Moorman, Parkin, et al. 2010); and an average field emission factor of 0.07%, well below the 0.75% IPCC comparison (Bock, Coleman, and Easton 2018). Across five Illinois bioreactors, only a small fraction of removed nitrate was emitted as N_2O and modelled watershed-scale indirect emissions were *lower* with a bioreactor than without (Brunton et al. 2024). Methane and sulfate responses are more variable: some bioreactors are notable CH_4 sources (soil covers lower surface CH_4) (Brunton et al. 2024), yet a dual-chamber system's landscape-scale impact was a $< 1\%$ increase in CO_2e, with pollution swapping more likely above 50-h HRT (Hartfiel, Hoover, et al. 2023). Sulfate reduction under non-N-limiting conditions is common, increases with HRT, and occurred in 83% of Illinois runs (Bell et al. 2015; Povilaitis and Matikiene 2020; Rivas et al. 2020). DOC leaches mainly at start-up as an initial flush that then

declines (Abusallout and Hua 2017; Rivas et al. 2020; Greenan, Moorman, Parkin, et al. 2009); it is dominated by high-molecular-weight fractions that are disinfection-by-product precursors but is removable by steel-chip filters or coagulation (Abusallout and Hua 2017).

8.4 Reactive media for phosphorus removal

Because woodchips remove little phosphorus—and often leach it at start-up—dedicated reactive filter media have been developed to retain the dissolved reactive phosphorus (DRP) that moves through drains in solution (L. Christianson, Cooke, et al. 2021). The evidence converges on a two-family model: iron- (Fe-)based media (iron oxyhydroxides, zero-valent iron, iron filings, steel by-products) retain phosphate mainly by strong inner-sphere adsorption, whereas calcium- (Ca-)rich media (blast-furnace and steel slag, limestone, lime-amended pellets) retain it by Ca-phosphate precipitation driven by dissolved Ca^{2+} and elevated pH; sorption is the preferred edge-of-field mechanism, and Fe-based media are generally more suitable because they bind phosphate stably over a wider pH range (Mendes et al. 2022; Z. Li and Arai 2020).

8.4.1 Media classes and mechanisms

Iron-based materials are the best-performing class, routinely removing $\geq 95\%$ of phosphate in batch and $> 90\%$ in most columns with negligible desorption (Allred and Racharaks 2014; Allred 2012; Allred, Martinez, and D. Gamble 2017; Mendes et al. 2022). Iron filings reached the highest capacity of 4.95 mg g^{-1} and steel slag 1.68 mg g^{-1}, versus only 0.10–0.13 mg g^{-1} for natural minerals—an order-of-magnitude advantage for industrial by-products (Sellner et al. 2019). Granular iron-oxide media retained SRP below detection far longer than steel by-products (Hauda, Safferman, and Ghane 2020), Fe-based CFH media were most robust under variable flow (Pugliese, Canga, et al. 2023), and iron-activated alumina reached 94.4% removal (Kumari and Dong 2025). Ca-rich slag and cement-kiln blends achieve $> 98\%$ removal at low-to-moderate loading (Agrawal, King, Fischer, et al. 2011; Sanford and Larson 2016), and lime-amended fly-ash pellets reached 1.98 mg P g^{-1} (S. Li, Cooke, L. Wang, et al. 2017), but their performance is governed by a finite supply of reactive Ca^{2+}: phosphorus is removed in a moving front as Ca^{2+} and pH-buffering capacity are consumed, downstream slag being more efficient than upstream (L. Wang et al. 2020), and field removal collapses once treated-water pH falls below 8.5, with 100% removal sustained only to $\sim$ 500 mg P kg^{-1} slag (Penn, Livingston, et al. 2020). Consistent with the two mechanisms, lower pH favours Fe-based adsorption whereas higher pH favours Ca-based precipitation (Mendes et al. 2022), and finer particle sizes and coexisting-ion chemistry (sulfate, bicarbonate, and DOC reduce uptake) modulate performance (Z. Li and Arai 2020; Sellner et al. 2019; Pugliese, Canga, et al. 2023). Natural minerals such as zeolite give lower, modest removal (21–58%) unless modified (Agrawal, King, Fischer, et al. 2011; Sadeghi et al. 2021).

8.4.2 Kinetics, longevity, and field structures

Phosphate uptake is rapid, permitting short contact times—10–15 s sufficed for near-complete removal from spiked drainage (Allred, Martinez, and D. Gamble 2017)—and Fe-based media exhibit low desorption and are regenerable, with a NaOH or KOH flush restoring capacity (Allred and Racharaks 2014; Allred, Martinez, and D. Gamble 2017; Kumari and Dong 2025; Sellner et al. 2019). Field structures confirm both feasibility and the laboratory pH and retention-time limits: a tile-drain structure removed 19.2 kg of dissolved P (55% of input) (Penn, Livingston, et al. 2020), an aluminium-treated steel-slag ditch structure removed $\sim$ 27% of dissolved P and 50% of total P (Shedekar, Penn, et al. 2020), and a compact activated-alumina structure treating the entire flow

removed 30% of the dissolved-P and 29% of the nitrate-N load (Penn, Williams, et al. 2024). Field removal typically underperforms the design basis (Shedekar, Penn, et al. 2020), and not all trials succeed—a Texas filter blend failed to reduce phosphate (while removing chlorothalonil) (Agrawal, King, Moore, et al. 2011). Coupling P media with woodchips addresses the woodchip P deficit: placing iron both upstream and downstream of woodchips enhances nitrate and orthophosphate removal (Goodwin, Bhattarai, and Cooke 2015; Hua et al. 2016; S. Li, Cooke, Huang, et al. 2018; Oladeji et al. 2023), and mixed-media bioreactors that initially release P become sinks within a year, cutting SRP 97% (Husk, Sanchez, et al. 2018; Z. Zhang et al. 2024). The central caveat is hydraulic: a stacked woodchip bioreactor removed 93% of the nitrate entering it, but because 93% of tile discharge bypassed it, this was only a 5.4% load reduction—whereas a P-removal structure treating the full flow removed nearly ten times more nitrate (Penn, Williams, et al. 2024).

8.5 Saturated buffers, denitrification walls, ditches, and in-channel practices

In-channel and edge structures add interception where drains discharge. Saturated riparian buffers, which redistribute tile flow laterally through the riparian soil, offer the largest climate-mitigation potential per unit area among conservation-drainage practices (Hall et al. 2025). Riparian zones remove nitrate efficiently by denitrification—groundwater nitrate fell from 7–8 to < 0.1 mg L^{-1} within 16–47 m (JACOBS and GILLIAM 1985; Schilling, Streeter, Pierce, et al. 2022)—and redirecting tile flow into a buffer removed nitrate at 1.01 ± 0.07 mg N L^{-1} m^{-1}, with chloride confirming denitrification rather than dilution (Jaynes and Isenhart 2014; Carstensen et al. 2021). Across six saturated buffers, all removed nitrate from tile-outlet water (13–179 kg N yr^{-1} over 3.4–40.5 ha; 8–84% effectiveness) at a mean cost of US\$2.94 kg^{-1} N, making them simple, cheap, and quick to install (Jaynes and Isenhart 2019). Denitrification beds followed Michaelis–Menten kinetics with very low greenhouse-gas emissions (Ghane, Fausey, and Brown 2015). In-channel practices contribute additional reductions: a low-grade weir cut nitrate load 58% (Strock and Ranaivoson 2025), a two-stage ditch reduced TP load 40% (Hodaj et al. 2017), and blind inlets and subsurface fertiliser placement reduced DRP 65–72% (Kröger, Dunne, et al. 2013; R. Wilson et al. 2019; Sharpley et al. 2019). Stacking practices multiplies benefit: a combined drainage-management–bioreactor–saturated-buffer system reduced nitrate export 78–88% (G. Singh, M. Davis, et al. 2024). A key limit is that where tile drains route water directly to streams they bypass the riparian zone, curtailing natural attenuation (Puckett and Hughes 2005).

8.6 Comparative effectiveness, cost, and watershed-scale integration

8.6.1 Load-reduction benchmarks and cost

Comparative syntheses give consistent benchmarks: denitrifying bioreactors, wetlands, cover crops, rotation, and nutrient management each reduced subsurface nitrate loads 23–40% (Y. Yuan et al. 2022), and nine BMPs in the Lower Mississippi Alluvial Valley delivered 15–100% nutrient reductions (Kröger, Perez, et al. 2012). A US Corn Belt review of 119 records found controlled drainage, water-quality wetlands, bioreactors, drainage water recycling, and saturated buffers all effective for nitrate but with differing co-benefits: wetlands gave the highest nitrate reduction plus P removal and habitat value, drainage water recycling the greatest yield benefit, and bioreactors effective nitrate removal but limited extra benefits (M. Mitchell et al. 2023). Global synthesis places bioreactor flow-season removal at 14–66%, wetlands near 39% TN, and buffers and controlled drainage at $\sim$ 30–50% annual N-load reduction (H. Wang and Ishii 2025). On cost, wetlands rank most cost-effective (2.04–2.91 \$ kg^{-1} N), followed by two-stage ditches and cover crops (Roley et al. 2016; Messer et al. 2021); bioreactor installation runs <\$5,000–27,000 at <\$2.50–$\sim$\$20 kg^{-1} N,

with corncob media, pumped systems, and spring bioreactors cheaper still (L. Christianson, Cooke, et al. 2021; Law, Slade, et al. 2023; Hartfiel, Díaz-García, et al. 2024; Easton, Bock, and Stephenson 2019). For phosphorus, edge-of-field practices retain P chiefly by particulate sedimentation (27–100%) but depend strongly on residence time and can flip from sink to source during events, so effective P control requires matching practice to site hydrology, connectivity, and legacy P (Mendes 2020; Kröger, Dunne, et al. 2013; G. Singh, G. Kaur, et al. 2020; King, Williams, Macrae, et al. 2015; Ni, Y. Yuan, and W. Liu 2020).

8.6.2 Scale attenuation, required reductions, and practice-stacking

Because tile drainage is the primary pathway controlling watershed water and nitrogen export, downstream targets translate to large required reductions at the tile outlet—about 55% for one Iowa river (Schilling, Jones, et al. 2012; Ikenberry, Soupir, Schilling, et al. 2014). Concentrations decline with drainage area as export dilutes downstream, and substantial in-network retention (34–82%) occurs via underground denitrification, though in-stream processing immediately below tile outlets offers little potential because its timing is poorly matched to seasonal loading (Schilling, Jones, et al. 2012; Billy, Birgand, Sebilo, et al. 2011; Billy, Birgand, Ansart, et al. 2013). This scale attenuation of in-field effects means edge-of-field benefits measured at a single tile can be diluted at the watershed outlet, and meeting large goals requires *combining* in-field and edge-of-field practices. In the Mississippi basin, nitrogen-management practices alone were insufficient, but N-management plus N-removal practices reached the 45% target while converting under 6% of cropland when targeted (McLellan et al. 2015), and LiDAR-based placement achieved 34–39% nitrate reduction while removing $< 5\%$ of cropland (Tomer, S. Porter, et al. 2015; Kvítek et al. 2023). Long-term records confirm that stacked BMPs work, with Czech outlet nitrate declining over 2004–2018 and combined practices cutting load 60.3% (Fraindová et al. 2022; Law, Long, et al. 2022). Crucially, traditional surface-water BMPs can be inadequate over tile drainage: seven years of strip-till, grassed waterways, and stream buffers produced no measurable change in nitrate, phosphorus, sediment, or hydrology because they did not intercept subsurface flow, which must therefore be a design priority (Lemke et al. 2011). Finally, monitoring design strongly affects inferred loads: high-frequency sampling captures storm dynamics that grab sampling misses, and significant nitrate declines are far more detectable in flow-weighted concentration than in load or yield (Drake et al. 2018; Danalatos et al. 2022; Irvine et al. 2019).

9 Other Water-Quality Dimensions: Pesticides, Microbes, Trace Elements, and Greenhouse Gases

Beyond nitrogen and phosphorus, subsurface drainage governs the field-to-stream transport of pesticides, manure-borne microbes and resistance genes, trace elements, and dissolved greenhouse gases, and it reshapes agricultural ecology. Across these dimensions the evidence converges on a common logic: the same preferential-flow pathways and hydrologic controls that dominate nutrient loss also govern contaminant delivery, and management—application method, timing, and edge-of-field engineering—frequently matters more than drainage per se.

9.1 Herbicides, Pesticides, and Emerging Contaminants

The timing of runoff and drainage relative to application is the single most powerful control on peak pesticide concentrations and losses. Synthesising more than thirty North American studies,

Kladivko, Brown, and J. Baker (2001) established that pesticides are commonly detected in drainflow during the first drainage event(s) after application, concentrations decline rapidly thereafter, and most mass losses are very small—generally less than 0.5% and often below 0.1% of applied—while tile concentrations and losses are usually much lower than in surface runoff, frequently by an order of magnitude. Site-specific values corroborate this: Illinois long-term atrazine losses were 0.11–0.22% of application (Kalita et al. 2006); New York 2,4-D and atrazine losses were 0.08% and 0.02% (Shalit and Steenhuis 1996); and less than 1% of applied herbicide was recovered in tile drains over two years (Fortin et al. 2002; Siimes et al. 2006). Early events dominate: most atrazine and metolachlor loss occurred within the first 61 days (SOUTHWICK et al. 1990), and over 80% of catchment losses occurred during just two rain events (Leu et al. 2004). In cold regions most loss can occur during snowmelt and thaw rather than immediately after application (Siimes et al. 2006), and two-phase seasonal behaviour is common (Novak, Portal, and Schiavon 2001; Dousset et al. 2004).

The mechanism reconciling near-immediate arrival with small overall losses is preferential (macropore) flow. Tracer breakthrough curves were typical of preferential flow, with herbicides arriving after less than 1 cm of net drainage, interpreted as transport through only $\sim 3\%$ of pore volume (Fortin et al. 2002; Dousset et al. 2004). Soil structure, rather than sorption or degradation, dominated between-soil differences, with greater metolachlor losses in a clayey Vertic Cambisol than a loamy Luvisol (Novak, Portal, and Schiavon 2001; Novak, Banton, and Schiavon 2003). Although mass losses are small, transient concentrations can exceed thresholds: Ontario maxima reached 287 μg L^{-1} (atrazine) in tile discharge, with the atrazine MCL exceeded in 60% of tile samples (GAYNOR, MACTAVISH, and FINDLAY 1995), and Iowa subsurface drains had glyphosate maxima above 250 μg L^{-1} (Coupe, Kalkhoff, et al. 2012). Degradates contribute meaningfully and can be more mobile than parents: deethylatrazine accounted for 10–27% of combined loss (GAYNOR, MACTAVISH, and FINDLAY 1995; MUIR and B. BAKER 1978; JAYACHANDRAN et al. 1994). Management levers include banding, residue-conserving tillage with intercrops (reducing loss 46–49%), and application-timing restriction via a soil-wetness index (Gaynor et al. 2001; Julien et al. 2023; Moorman, Jaynes, et al. 1999), whereas corn-stover harvest can worsen drainage quality (Shipitalo, Malone, et al. 2016). Woodchip and biochar-amended bioreactors removed atrazine (up to 93%) mainly abiotically without inhibiting denitrification (Hassanpour et al. 2019; Ilhan, Ong, and Moorman 2011), and modelling frameworks (PESTDRAIN, HYDRUS-2D, RZWQM, SWAT+) reproduce drainage pesticide dynamics when macroporosity and sorption kinetics are represented (Branger et al. 2009; Boivin et al. 2006; Wendell et al. 2024; Novak, Banton, and Schiavon 2003). Tile drainage also transports emerging contaminants—PPCPs, hormones, PBDEs, perfluoroalkyl acids, and neonicotinoids—whose behaviour mirrors the timing and preferential-flow patterns above, generally at low, transient concentrations (Gottschall, Topp, Metcalfe, et al. 2012; Gottschall, Topp, Edwards, Payne, Kleywegt, Russell, et al. 2013; Gottschall, Topp, Edwards, Payne, Kleywegt, and Lapen 2017; Ghirardini and Verlicchi 2019; Schaafsma et al. 2019).

9.2 Manure-Borne Microbial Contaminants and Resistance Genes

Manure and organic amendments introduce fecal microorganisms, resistance genes, hormones, and trace elements that drainage can rapidly route to receiving waters, with application method and soil moisture—rather than manure presence alone—largely governing transport. Curli-producing *E. coli* were far more abundant in tile drains from farms that surface-applied manure than one that incorporated it, indicating that application method itself shapes the transported phenotype (Truhlar et al. 2015). Tillage governs pathogen delivery mainly through macroporosity: zero tillage generally produced higher subsurface flow and greater *E. coli* loads than conventional tillage (Thi-

agarajan et al. 2007; Jenkins et al. 2009). Antecedent soil moisture and water-table position exert first-order control: effluent on wet soil produced drain-flow *E. coli* of 10^5–10^7 per 100 ml versus $\sim 10^4$ under a greater moisture deficit, and where preferential flow occurred, drainage concentrations approached those of the applied effluent (Collins et al. 2007; Ding et al. 2021). Manure is also a reservoir of antibiotic resistance genes that persist through drained soils into drainage water: shotgun metagenomics identified 320 unique ARGs, with manure treatment shifting soil and drainage-water resistomes and many manure-derived genes associated with mobile genetic elements in food- and water-borne pathogens (S. Smith et al. 2019). Steroid hormones were detected only transiently and generally did not differ significantly among treatments (Jenkins et al. 2009). Engineered receiving features can attenuate microbial loads—a constructed wetland reduced nitrate and *E. coli* mass loads 67.6% and 63.3% (Haverstock et al. 2017)— though some wetlands act as net *E. coli* exporters (Stott et al. 2023). Drained soils, bioreactors, and wetlands host distinct, dynamic microbial communities structured by depth and season, and favourable conditions can be engineered to develop effective denitrifier assemblages (Andrus et al. 2014; M. Porter et al. 2015; Hathaway, M. Porter, et al. 2015; Knorr, Weisener, and Phillips 2023; Schilling, Kult, et al. 2017).

9.3 Trace Elements, Selenium, and Acid Sulphate Hazards

Where drainage concentrates trace elements, it can create acute ecological hazards. The Kesterson Reservoir remains the definitive case: waterborne selenium averaged 0.122 mg L^{-1} ($\sim$ 230 times reference), biomagnified through the food chain, and caused dead and deformed bird embryos, with integrated studies fulfilling the equivalent of Koch's postulates and identifying selenium as the sole causative agent (SAIKI and LOWE 1987; Ohlendorf 2002; BENSON, DELAMORE, and HOFFMAN 1993). The regulatory legacy is durable: open-water evaporation is now prohibited in California, requiring rock-media systems (Ayars and Soppe 2014; BRADFORD et al. 1991), and selenium-removal wetlands retained about 59% of inflow Se (S. Gao, Tanji, et al. 2003). In evaporation basins, evapoconcentration accumulates extreme salinity and trace elements (WESTCOT 1988), and redox chemistry jointly controls arsenic and uranium mobility (Ryu, S. Gao, and Tanji 2010; S. Gao, Ryu, et al. 2007; Duff, Amrhein, and Bradford 1997); boron isotopes and other tracers can distinguish contaminant sources (Tayfur, Tanji, and Baba 2010; Komor 1997; X. Zhou, X. Zhao, et al. 2023). At the opposite extreme, intensive drainage can drive negative micronutrient balances and net soil depletion (Gul et al. 2013), while central-Iowa drainage ionic composition was dominated by bicarbonate, calcium, and magnesium (Zimmerman and Kaleita 2017). Drainage of acid sulphate soils is a distinct hazard: oxidation of pyritic sediments drives stream pH declines to near or below 4.0 and elevates aluminium and metals, with controlled pipe drainage recommended over in-stream liming (Osterholm, Åström, and Sundström 2005; Åström et al. 2005; Virtanen, Puustinen, and Yr-Hall 2017; Salo, Virtanen, et al. 2023).

9.4 Greenhouse-Gas Dimensions

Drainage nitrate is coupled to greenhouse-gas fluxes, and drainage water is commonly supersaturated with dissolved N_2O, CH_4, and CO_2. In Japan, indirect emissions via subsurface drainage were equivalent to 4.6%, 58.3%, and 2.5% of direct emissions, with N_2O-N/NO_3-N ratios below the IPCC factor (Sawamoto et al. 2003; Minamikawa, Nishimura, et al. 2010; Minamikawa, Eguchi, et al. 2013). In Canada, dissolved N_2O losses approached surface emissions, were strongly seasonal, and declined along tile length, implying an in-drain sink and likely underestimation of the IPCC coefficient (Burton, Wilts, and Macleod 2024); a global synthesis found subsurface drains had the highest mean emission factor among artificial waters, with the stationary IPCC factor tending to

overestimate N_2O production (Webb, Clough, and Quayle 2021). Conversely, installing drainage sharply reduces methane in paddy systems by improving aeration: drained fields emitted 71% less CH_4 than non-drained fields (Shiratori et al. 2007; Furukawa, Shiratori, and Inubushi 2008). Management modulates N_2O: controlled drainage kept cumulative N_2O below 1% of applied N, and saturated riparian buffers removed nitrate without increasing N_2O (E. Smith, Vosman, et al. 2019; M. Davis et al. 2019; R. Nash et al. 2015; Carstensen et al. 2021). Carbon export from drained fields is dominated by inorganic carbon through tiles ($\sim$ 96% of total carbon export), so reducing water throughput supports carbon retention (Schilling, Streeter, Jones, et al. 2023; Driscoll et al. 2025).

9.5 Ecological Responses to Subsurface Drainage

Drainage reshapes agricultural ecology from soil fauna upward. *Lumbricus terrestris* abundance was typically higher above drains than between them (median counts twice as high, fresh mass five times), so drainage pattern affects population density and should inform sampling design (Nuutinen, Pöyhönen, et al. 2001; Nuutinen, Butt, and Jauhiainen 2011). Drainage type is associated with farmland-bird outcomes, with skylark and meadow pipit favouring open-drained over subsurface-drained fields (Marja et al. 2013), while tile-drainage wetlands carried cumulative pesticide concentrations 4.7 times higher than reference wetlands (neonicotinoid-dominated), with measurable neurological effects in juvenile amphibians (Campbell et al. 2023). Subsurface drainage of meadows was a leading cause of decline in semi-aquatic snails (Dreyfuss, Vignoles, and Rondelaud 2018). Restored aquatic habitats can benefit: in restored Iowa oxbows, tile drainage had negligible effects on physical or biotic variables, providing promising evidence that multipurpose oxbows can enhance biodiversity without ecological penalty, with outcomes improved by sediment excavation (Leberg et al. 2024; Winikoff et al. 2020; Harris et al. 2018). Crucially, management practices often exert clearer effects than tile drainage itself: no-till and diverse rotations altered bulk density, organic matter, and aggregate stability while cover crops and drainage had minimal effect in some settings (Saha et al. 2024; Hermawan and Bomke 1996; Sherbine et al. 2023), and additional practices such as polyacrylamide amendment and anti-weed mesh with drainage provide further quantified nutrient-retention and production benefits (Jiang et al. 2010; Giménez-Valero, Maciá-Vázquez, et al. 2024; Giménez-Valero, Núñez-Gomez, et al. 2025).

10 Agronomic Benefits: Yield, Salinity Reclamation, and Socio-Economics

Beyond its hydrologic and water-quality functions, subsurface drainage is a production technology whose agronomic and economic value is well established. The evidence—multi-year paired-field trials, long-term monitoring, and calibrated simulation across humid, arid, monsoonal, and cold-climate settings—supports four consolidated conclusions: drainage and its managed variants raise crop yields, most strongly on poorly drained soils and in wet years, while maintaining or improving water-use efficiency; subsurface systems reliably leach salts and reclaim saline and waterlogged land; saline and brackish water can be reused safely where salt balance is maintained; and drainage investment delivers favourable financial and social returns. Meta- analyses anchor the headline magnitudes: a global synthesis of 891 experiments confirmed drainage significantly reduces soil salinity and raises yield (Y. Yang, Ao, et al. 2025), and a U.S. meta-analysis of 154 comparisons found controlled drainage and subirrigation raised yield 8.0% over free drainage (N. Singh et al. 2022). Spacing, depth, and outlet-control mechanisms are developed in the design section; here we

own the yield, reclamation, reuse, and socio-economic outcomes, together with the salinity-specific design optima and salt-balance dynamics.

10.1 Crop yield and agronomic outcomes

10.1.1 Magnitude of yield gains

Direct drained-versus-undrained comparisons repeatedly document substantial yield increases, primarily by relieving early-season waterlogging. Artificial drainage raised soybean yield 275 kg ha^{-1} (8%) in experiments and 157 kg ha^{-1} (4%) in producer fields (Mourtzinis et al. 2021), and in a long-term Indiana experiment corn yields rose progressively in all drained treatments across 24 corn years while the undrained control stayed flat, though soybean was unaffected by spacing (Rui, Goller, and Kladivko 2024; Kladivko, Willoughby, and Santini 2005). The largest relative gains occur in heavily stressed environments: Nile-Delta drainage doubled cotton and rice yields and raised wheat and clover 50% (ELMOWELHI et al. 1988); Indian acid saline "kari" soils gained 1.36 t ha^{-1} (61%) (Mathew, Panda, and Nair 2001); Bangladeshi waterlogged saline clay gained 30–64% (M. Islam et al. 2025; M. Islam et al. 2022); and an Alberta system raised barley grain yield from 1310 to 3900 kg ha^{-1} (BENNETT et al. 1982). More moderate but robust gains dominate temperate systems: shallow drainage in North Carolina raised corn 4% and soybean 14.3% (Watkins et al. 2024); Iranian paddies exceeded controls by 13.4–31.3% (up to 66–70%) (Darzi-Naftchali, Ritzema, et al. 2017; Darzi-Naftchali and Shahnazari 2014); and Chinese blind-ditch and pipe drainage raised rice 8.0% and 6.0% (B. Qi et al. 2025).

10.1.2 Controlled drainage, subirrigation, and recycling

Managed practices deliver benefits beyond free drainage when they retain water for crop use in dry periods—quantified at 8.0% average yield gain across 154 comparisons (N. Singh et al. 2022). In Missouri, free drainage and drainage water recycling raised corn yield 20% and 26% over a nondrained control (recycling reaching 48% in some years), avoiding drought losses (H. Kaur, Nelson, and G. Singh 2021); in North Carolina recycling raised corn and soybean 8% and 17% (Moursi, Youssef, and Poole 2024). Subirrigation is strongly site-dependent, governed by lateral hydraulic conductivity: across three Ohio sites it improved relative yield to 80–93% at two sites but had almost no effect where conductivity was very low, and simulations projected it increasingly outperforming free drainage under future climate (Gunn, Baule, et al. 2018; Gunn, Allred, et al. 2019). Controlled drainage and subirrigation kept soil water optimal in both wet and dry years (Niaghi et al. 2019), improved claypan soybean yield while reducing interannual variability and allowing earlier planting (Nelson, Smoot, and Meinhardt 2011), and combined with foliar pyraclostrobin produced synergistic soybean gains up to 1070 kg ha^{-1} (Nelson and Meinhardt 2011).

10.1.3 Mechanisms, water-use efficiency, and boundary conditions

The proximate mechanism is relief of excess-water stress and restoration of root-zone aeration (BENNETT et al. 1982; Darzi-Naftchali and Shahnazari 2014), with growth-stage sensitivity (cotton flowering and boll formation most sensitive; surface waterlogging worse than subsurface) (L. Qian et al. 2021). Improved aeration propagates through physiology—raising emergence, root vigour, chlorophyll, and dry matter in cotton (Y. Yang, X. Zhou, et al. 2022), lowering leaf Na^+/K^+ in sunflower (M. Islam et al. 2022), and raising rice nitrogen-use efficiency 16.1% (Darzi-Naftchali

and Mokhtassi-Bidgoli 2022). Controlled drainage repeatedly improves irrigation water-use efficiency without yield loss: simple management of existing Egyptian systems improved WUE 15–20% (Wahba, Christen, and Amer 2005), controlled drainage cut applied irrigation 14–16% (Mahmoud et al. 2021), and Chinese buried pipes raised water productivity, highest near 1.4 m burial (Y. Liu, Zeng, et al. 2025). Yield response is soil- and weather-specific: only undrained controls show negative rainfall responses, wet years favour narrower spacing and dry years wider (Rui, Goller, and Kladivko 2024; Kladivko, Willoughby, and Santini 2005; Acharya, Chatterjee, and Daigh 2019), and drainage advanced sowing 7–8 days and avoided 29–52% harvest population losses in wet years (Mourtzinis et al. 2021; Nelson, Smoot, and Meinhardt 2011). The evidence also delineates limits: where surface drainage is adequate, subsurface drainage did not raise grain yield (Wiersma et al. 2010); medium-textured soils are most responsive while fine and coarse show no clear effect (N. Singh et al. 2022); over-drainage can induce drought (Cordeiro and Ranjan 2015; DeBoer and Beck 2006); an optimum water-table depth exists that varies with soil, crop, and permeability (HOOGHOUDT 1952; Bigelow et al. 2001); and installation without maintenance underperforms (Malota and Senzanje 2016). Adoption is a leading adaptive practice—about three-fourths of surveyed U.S. producers report drained cropland (Morton et al. 2015), and nearly 90% of Finnish farmers rate subsurface drainage important, ahead of irrigation (Peltonen-Sainio, Sorvali, and Kaseva 2021).

10.2 Salt leaching and reclamation of saline and waterlogged soils

10.2.1 Reclamation efficacy

Subsurface drainage reliably lowers the water table below the root zone and removes salt from the upper profile, converting saline and waterlogged land into productive cropland, with topsoil salinity reductions commonly 35–80% and corresponding yield gains, validated at production scale (Y. Yang, Ao, et al. 2025; Derby et al. 2024). In India, drained-block salinity fell from 7.18 to 4.62 dS m^{-1} (mean 35.7%) while undrained soils rose, and yields increased 18.75–27.62% (D. Sharma and S. Gupta 2006); wide-spacing reclamation lowered topsoil salinity from 8.4 to about 2.1 dS m^{-1} and raised cropping intensity from 143% to 191% (Manjunatha et al. 2004). Chinese reclamations are similarly large: drip-plus-pipe cut EC from about 15 to 2.9–6.9 dS m^{-1} and raised seed-cotton yield 52% (Heng et al. 2021), and after four years the 0–1.0 m layer fell from 27.6 to 4.1–7.2 dS m^{-1} (C. Zhang et al. 2021). Desalinization is surface-concentrated—0–20 cm rates exceed 80% while deeper layers lag and the meta-analytic reduction is nonsignificant below 80 cm (D. Han et al. 2023; Y. Yang, Ao, et al. 2025). Comparable reductions are reported for Turkey, Tunisia, and Canada, the last raising barley yield up to twentyfold (Bahçeci and Nacar 2009; Haj-Amor and Bouri 2019; Steppuhn 2006; Steppuhn and McArthur 2017), and reclamation homogenises the field and improves porosity and infiltration (S. Yu, J. Liu, Eneji, H. Liu, et al. 2015; S. Yu, Cheng, et al. 2025; D. Sharma, K. Singh, and Rao 2000). Where the water table stays shallow, however, combined waterlogging and salinity are more damaging than waterlogging alone, with yields failing above 12 dS m^{-1} (Kahlown and Azam 2002).

10.2.2 Design parameters governing salt removal

Closer spacing and greater depth shorten the desalinization period and increase salt removal, though depth benefits saturate. A DRAINMOD analysis found salinity fell 55.5% in two years at 20 m spacing but only 5.9% in five years at 200 m, a slightly-saline target within a decade requiring spacing below 30 m (Y. Chen et al. 2025); the meta-analysis generalises this into climate-specific layouts (spacing $>$ 30 m and depth $>$ 1.2 m in humid climates; 10–20 m spacing in arid climates

with shallow water tables) (Y. Yang, Ao, et al. 2025). Field and column studies identify optima near 15–20 m (Z. Wang, Heng, et al. 2020; D. Han et al. 2023; G. Chen, Wei, and H. Liu 2022), with desalination decaying beyond about 30 m (Q. Zhu et al. 2025). Depth effects are pronounced but bounded—pipes shallower than 1.0 m are essentially invalid for discharging salt (Y. Qian, Y. Zhu, et al. 2021), yet returns diminish beyond about 1.4 m (Bahçeci and Nacar 2007; Bahçeci, Dinç, et al. 2006), and an intermediate 0.8–1.2 m depth is often preferred under saline irrigation, saving 10–15% water (Feng et al. 2017; Muhammad, Ibrahim, and El-Sayed 2021). Envelope and filter design is a first-order control: the envelope factor dominates pipe material, a double-layer geotextile achieving $> 95\%$ salt removal while triple layers clog (H. Li et al. 2022; X. Wang, Y. Zhang, et al. 2024; Elzoghby, Z. Jia, and Luo 2021). Economic analyses converge on roughly 20 m spacing at 1.2 m depth as the maximum-return layout (Y. Chen et al. 2025).

10.2.3 Salt balance, temporal decline, and leaching structure

Because tile drains intercept salt-laden water, effluent salinity is characteristically higher than irrigation-water salinity and declines over years as stored salt is depleted (H. Li et al. 2022; Karegoudar et al. 2022), with ECdw/ECiw ratios of 11–17 in Iran (Akram et al. 2013). Salt-balance accounting is the standard diagnostic: an oasis exported about 94% of imported salt while discharging 42% of irrigation water (Askri, Khodmi, and Bouhlila 2022), though clogging can cut removal to 15–27% (Ben Aissa et al. 2013; Haj-Amor and Bouri 2019). Most residual salt is removed within ten years, effluent-salinity–drain-age relationships being stable across decades (JOHNSTON 1993), though modeling indicates 15–50 years to full equilibrium, with drainage-water salinity more sensitive to leaching fraction than to spacing (Kelleners, Kamra, and Jhorar 2000). Salt export increases with both depth and spacing, depth having the larger effect where groundwater quality declines with depth (GRISMER 1993; KAMRA, S. SINGH, RAO, and VANGENUCHTEN 1991; KAMRA, S. SINGH, and RAO 1994). Leaching is vertically stratified (upper $>$ middle $>$ lower) and ionically selective, with Cl and Na leached most efficiently (91% and 88% at 0–20 cm) and Ca persisting as precipitated gypsum (G. Chen, Wei, and H. Liu 2022; Y. Qian, X. Han, Y. Zhu, S. He, et al. 2025; S. Yu, J. Liu, Eneji, L. Han, et al. 2016). Re-accumulation is common, following a "wash-out–recovery–accumulation" pattern with surface re-salinization in dry periods (Q. Zhu et al. 2025; Dou et al. 2022; Shi et al. 2021). Mechanistically, under complete ponding flow concentrates near the drain ($\sim 80\%$ of flow through 20% of soil volume), so progressive or partial flooding, low-permeability barriers, and surface mulches raise efficiency substantially (a 10 m barrier saved 52–82% of leaching water; a mulch cut effluent volume 81%) (RAO and LEEDSHARRISON 1991; Youngs and Leeds-Harrison 2000; X. Yu, Dan, and Xin 2018; J. Zhang, Werner, and C. Lu 2022), whereas macropore bypass and fine texture prolong leaching (Xin, X. Yu, et al. 2016; MANN, PISSARRA, and VANHOORN 1982).

10.3 Saline and brackish water reuse

Subsurface drainage enables safe reuse of saline and brackish water where salt balance is maintained by adequate leaching. Blended drainage water up to 9 dS m^{-1} produced about 90% wheat yield in India, cyclic use generally outperforming continuous use (D. Sharma and Tyagi 2004); five-year data gave relative wheat yields of 95.8, 90.3, 83.7, and 77.8% at 6, 9, 12, and 18.8 dS m^{-1} with root-zone salinity kept below 4 dS m^{-1} (D. Sharma and Rao 1998). In China, saline (6 g L^{-1}) irrigation matched freshwater wheat yields with continued salinity decline (H. Gao et al. 2023), and drainage kept saline irrigation viable up to 6.25 dS m^{-1} for maize (Feng et al. 2017) and 8.0 dS m^{-1} under cut-soiler drainage (Neha, G. Yadav, R. Yadav, A. Kumar, Rai, Sehwag, et al. 2022), with

cotton optima at 2–4 g L^{-1} (D. Li et al. 2024; P. Lu et al. 2019). Reuse nonetheless carries sodicity and accumulation risks: cyclic reuse caused profile salinity fluctuations and sodic surface soils (SARe > 11) (J. Mitchell et al. 2000; Jayawardane et al. 2001), though modeled reuse over subsurface drains caused only marginal groundwater deterioration (KARAJEH and TANJI 1994). Integration with complementary practices raises performance: combined pipe-plus-ditch systems reached 170% of a single ditch's efficiency (Yao et al. 2024); timing drainage to snowmelt more than doubled surface salt leaching (S. Yu, J. Liu, Eneji, L. Han, et al. 2016); cut-soiler shallow drainage cut salinity 50–60% (Neha, G. Yadav, R. Yadav, A. Kumar, Rai, Sehwag, et al. 2022; Neha, G. Yadav, R. Yadav, A. Kumar, Rai, Onishi, et al. 2022); and regional coupling of optimised drip irrigation with tailored drainage reduced district-wide irrigation 10% and salinity 15% (Y. Liu, Ao, et al. 2026). At basin scale, persistent waterlogging and salinity in Central and South Asia reflect deteriorated infrastructure and inadequate effluent disposal, underscoring that drainage must be managed within an integrated salt balance (Dukhovny et al. 2007; Qureshi et al. 2008; Ritzema 2016).

10.4 Socio-economic returns and investment viability

10.4.1 Yield, income, and productivity gains

Drainage investment generates multi-dimensional returns—higher yields, cropping intensity, farm and household income, land value, and employment (Gopalakrishnan and Kulkarni 2007; Abdel-Dayem, Abdel-Gawad, and Fahmy 2007). Drainage typically raises average yields 10–30% and lifts net farm income where leaching water and sustainable disposal are assured (Abdel-Dayem, Abdel-Gawad, and Fahmy 2007; Datta, Jong, and O. Singh 2000; Wichelns 2002). In the Nile Delta, net incomes rose by 180–375 per hectare with yield gains accruing progressively from 50% in year one to 100% by year four (Wichelns 2002). In India, the Uttar Pradesh project reclaimed over 175,000 ha of sodic land, raising farm income over 150%, cropping intensity from 49% to over 200%, and household income 53% (Gopalakrishnan and Kulkarni 2007), and Haryana pilots raised cropping intensity and wheat yields as salinity fell from about 50 to 5 dS m^{-1} (Datta, De Jong, and O. Singh 2002; Datta, Jong, and O. Singh 2000). Long-term drainage also improves the soil resource base, raising the soil quality index 76–136% with carbon- sequestration co-benefits (Mukhopadhyay, Fagodiya, Prajapat, et al. 2023; Mukhopadhyay, Fagodiya, Narjary, et al. 2023), and in Pakistan a project raised cropping intensity from 102% to 157% and living standards 30% (Ghumman et al. 2011).

10.4.2 Financial appraisal, scale, and cost-reducing design

Formal appraisals confirm favourable performance. The RAJAD project (14,925 ha) reported a benefit–cost ratio of 1.28–2.87, payback of 4.5–6.9 years, and an internal rate of return of 18–35% (Gopalakrishnan and Kulkarni 2007); Nile-Delta drainage gave farm-level benefit–cost ratios of 10.3 (non-saline) and 5.5 (saline) (Wichelns 2002), and small-scale manually installed drainage yielded IRRs of 8–11.5% where water for leaching and disposal are available (Datta, Jong, and O. Singh 2000; Datta, De Jong, and O. Singh 2002), though not all appraisals are high (Ghumman et al. 2011). National programmes reflect this importance: Egyptian drainage serves about 2.4 million hectares (Abdel-Dayem, Abdel-Gawad, and Fahmy 2007), India reclaimed about 1.39 million ha (Gopalakrishnan and Kulkarni 2007), and globally 12–17% of cultivated area is drained (Valipour 2014). Careful design sharply reduces cost: Nile-Delta design charts gave minimum network costs of 205–374 per hectare at 300 m lateral length (37% below 100 m) (Gabr, Fattouh, and Eltarabily 2024); corrugated plastic pipe and factory pre-wrapping of envelopes reduced installation costs by

roughly 50% (Yannopoulos et al. 2020); and mechanization with laser grade control and improved envelopes lowered costs and raised quality (Ritzema, Nijland, and Croon 2006; Ritzema, Wolters, et al. 2007; TeKrony, Sanders, and Cummins 2004). Disposal infrastructure follows economies of scale, with economic outcomes strongly conditioned by disposal costs (J. Singh and Christen 2001; Bethune, Q. Wang, and Gyies 2008; HANSON and FULTON 1994; Napierala 2024).

10.4.3 Policy and externality context

Because drainage water carries disposal costs and externalities, its economics interact with policy. In California an unregulated regime produced applied-water and drainage volumes above efficient levels, whereas a direct drainage fee induced efficiency (net social benefit 1132 US$/ha) (DINAR, KNAPP, and LETEY 1989). Experience underscores that economics alone do not govern outcomes: management must be technically sound, economically viable, and socially acceptable (Letey 1999), and basin-wide accounting shows combination strategies can cut soil salinity and river salt loading while raising yields (Burkhalter and Gates 2006; Quinn 2009; McDowell, Monaghan, et al. 2021). Because artificial drainage shifts diffuse phosphorus toward the subsurface pathway (Lucci et al. 2014; Allaire et al. 2011; Dadfar et al. 2010), and because deeper, wider drains raise drain-water salinity and salt load, drainage water quality is increasingly advanced as a necessary design criterion alongside production, with active water-table management and decision support the key future directions (Ayars and Evans 2015; Yannopoulos et al. 2020; Ritzema, Abdel-Dayem, et al. 2023).

11 Modeling, Measurement, and Mapping of Drainage Systems

The physical findings summarised in the preceding sections rest on a mature toolkit for simulating, measuring, and locating drainage systems. Across site-scale validations, benchmark comparisons of analytical solutions, pipe-scale experiments, watershed representations, formal calibration and uncertainty analyses, and geospatial detection, a robust set of consolidated findings has emerged: process models reproduce drainage flow and nitrate reliably after calibration but with skill that improves strongly with temporal aggregation; treating drainage as purely saturated flow over-predicts discharge; saturated conductivity and evapotranspiration dominate model sensitivity; data-driven and remote-sensing methods now complement process models; and no single geophysical or imaging technique is universally best for mapping buried networks. This section consolidates that methodological evidence so the physical and management conclusions can stand without model mechanics.

11.1 Field-scale drainage and water-quality models

DRAINMOD is by far the most extensively tested field-scale water-management model and is repeatedly judged a reliable design and evaluation tool for poorly and artificially drained lands, its process-based structure with independently measurable inputs describing drainage hydrology across many climates when soil hydraulic properties are adequately characterised (A. Singh 2019; R. Skaggs, Youssef, and Chescheir 2012). A defining, cross-cutting result is that skill is scale-dependent in time: water-table depth and drainage are reproduced with accuracy improving markedly from daily to monthly to annual scale. In North Carolina, water-table and daily-drainage Nash–Sutcliffe efficiencies were 0.49–0.73, monthly drainage 0.77–0.87, and annual volumes within 5% of observations (R. Skaggs, Youssef, and Chescheir 2012); Iowa and eastern-Canada cumulative drainage fell within 2–10% and about 30 mm of measurements (R. Singh, Helmers, and Z. Qi 2006; Dayyani,

Madramootoo, Enright, et al. 2009). The persistent weak point is daily peak flow, which is systematically underestimated (Morrison, Madramootoo, and Chikhaoui 2014; P. Singh et al. 1996), and uncertainty widths grow from daily to yearly scale while annual predictions remain within bounds (X. Wang, Frankenberger, and Kladivko 2006). The model has been validated well beyond its humid origin—cold Canadian snowmelt (R^2 up to 0.93), semi-arid India, and Illinois claypan soils (Dayyani, Madramootoo, Enright, et al. 2009; G. GUPTA et al. 1993; MOSTAGHIMI, MCMAHON, and LEMBKE 1989; Abduljaleel et al. 2023; Awad, El-Rawy, et al. 2022)—but delineated failure modes include unaccounted deep and lateral seepage (Sinai and Jain 2006), absent time-variable depression storage (Abduljaleel et al. 2023), and its parallel-drain geometry requiring plot-by-plot calibration in irregular landscapes (X. He et al. 2002). Where site data are scarce, pedotransfer estimation (ROSETTA, van Genuchten fits) with automated calibration (PEST) drives it reliably (R. Singh, Helmers, and Z. Qi 2006; Z. Qi, R. Singh, et al. 2015; Malota, Taguta, et al. 2024; Malota and Senzanje 2015), and enhancements target the daily-peak weakness through macropore flow (capturing ∼75% of peak events) and neural-network surrogates (Askar, Youssef, Chescheir, et al. 2020; Awad, Luo, and El-Rawy 2022; Awad, El-Rawy, et al. 2022); the evapotranspiration algorithm is a pronounced sensitivity, the best ET method not necessarily best for water-table prediction (Malakshahi, Darzi-Naftchali, and Mohseni 2020).

Richards-equation and comparative codes generally perform comparably. HYDRUS (2D/3D) predicts fluxes and water tables closely (EF 0.91–1) and clarifies mechanisms such as conductivity control of drawdown in compacted soils (Karandish, Darzi-Naftchali, and Simunek 2018; Darzi-Naftchali, Karandish, and Simunek 2018; Mante, Ranjan, and Bullock 2018; Guedessou et al. 2021; Ebrahimian and Noory 2015); CSUID, WAVE, SWAP, WaSim, CATHY, RZWQM, WEPP, ADAPT, and the few-parameter SIDRA family have all calibrated satisfactorily, with DRAINMOD often giving the most accurate discharge among them (Alzraiee, Garcia, and Gates 2013; Alzraiee, Gates, and Garcia 2013; El-Sadek, Feyen, and Berlamont 2001; Prasher et al. 1996; A. Verma, S. Gupta, and Isaac 2014; Malota, Mchenga, and Chunga 2022; Muma, Rousseau, and Gumiere 2017; Malone, Nolan, et al. 2014; Xian et al. 2017; Revuelta-Acosta et al. 2021; Mendez et al. 2004; Zimmer, Tournebize, et al. 2023; Henine, Jeantet, et al. 2022; LESAFFRE and ZIMMER 1988).

For water quality the same pattern recurs across the model families (DRAINMOD suite, RZWQM/RZWQM2, SWAT/APEX, APSIM, DNDC, HYDRUS): after calibration they reproduce annual and cumulative drainage and nitrate loss well—frequently within ±5–20%—and capture the direction of management effects, but predictive skill falls with temporal disaggregation, nitrate is predicted less accurately than flow, and daily nitrate concentrations, crop N uptake, and residual soil N are recurrent weak points (Youssef et al. 2018; Negm, Youssef, R. Skaggs, et al. 2014; S. Singh, Bhattarai, et al. 2020; Dayyani, Prasher, et al. 2010; L. Ma, Ahuja, et al. 2001; H. Liang et al. 2018; Ikenberry, Soupir, Helmers, et al. 2017; Marjerison et al. 2016; Ojeda et al. 2018; J. Qi et al. 2025; L. Ma, Malone, Heilman, Karlen, et al. 2007; Ahuja and Hatfield 2007). DRAINMOD-N II and RZWQM2 both satisfactorily predict tile nitrate, and DRAINMOD-DSSAT predicts drainage, nitrate, and yield very well (Du et al. 2017; Z. Qi, L. Ma, et al. 2012; Negm, Youssef, and Jaynes 2017; S. Singh, Negm, et al. 2022); cropping-system models (DSSAT, RZWQM, DNDC, APSIM, SWAT) likewise reproduce cover-crop and rotation effects on nitrate loss (Jeong and Bhattarai 2018; Ahmed et al. 2007; Ale et al. 2012; Pasley et al. 2021; Motarjemi et al. 2021). Phosphorus extensions are advancing but remain limited: RZWQM2-P first simulated daily dissolved reactive P and its pathway partitioning, while DRAINMOD-P predicted discharge and total P but poorly predicted DRP, and few standard models yet represent tile P (Sadhukhan, Z. Qi, T. Zhang, Tan, L. Ma, and Andales 2019; Grewal et al. 2025; Bin Shokrana, Ghane, and Z. Qi 2023; Askar, Youssef, Vadas, et al. 2021; Askar, Youssef, King, et al. 2025; Radcliffe et al. 2015; Z. Wang,

T. Zhang, et al. 2020). Cold-season snowmelt and freeze–thaw errors motivated dedicated developments (DRAINMOD 5.1, DRAIN-WARMF, RZ-SHAW), and calibration studies repeatedly identify lateral saturated conductivity in the tile layer and field capacity as controlling predictions (S. Wang et al. 2006; Dayyani, Madramootoo, Prasher, et al. 2010; Z. Li, Z. Qi, Madramootoo, et al. 2024; L. Ma, Malone, Heilman, Ahuja, et al. 2007; Malone, Kersebaum, et al. 2017).

11.2 Flow physics: saturated and unsaturated flow and preferential-flow representation

A central mechanistic result is that treating drainage as purely saturated flow overstates discharge. A saturated-flow model over-predicted seepage and drainage time by roughly an order of magnitude relative to experiment, an analytical solution, and the SUTRA Richards-equation model, because a vadose zone retaining substantial water persists even at steady state (Xin, Dan, et al. 2016a). Saturated–unsaturated drainage is governed by gravitational, pressure, capillary, and air-pressure potentials, so saturated conductivity alone is an inadequate design basis (Y. Li et al. 2021). Field diagnosis in Japanese paddies showed that vertical, not horizontal, resistance limits the maximum drainage rate (Nishida et al. 2020), accounting for vertical head loss improved water-table prediction (Talukolaee, Shahnazari, and Ritzema 2022), and retaining unsaturated-zone buffering matched numerical models closely whereas ignoring it distorted transients (Van de Craats, Van der Zee, and Leijnse 2021); equivalent-medium representations reproduce integrated outflow well but can misrepresent internal dynamics where fast vertical flow occurs (Carlier, Kao, and Ginzburg 2007). At field-to-hillslope scale the representation of macropores is decisive for reproducing rapid drainage. Dual-permeability and dual-porosity models reproduce hydrographs well ($R^2 = 0.84$; groundwater $r = 0.98$) while single-porosity models reach acceptable accuracy with far fewer parameters, though all fail to match low dry-season discharge (Varvaris et al. 2018); dual-permeability formulations enable low matrix conductivity while still reproducing fast events (Warsta et al. 2014; Salo, Warsta, et al. 2017; Koivusalo, Karvonen, and Paasonen-Kivekäs 1999), adding macropore flow improves peak simulation without changing total drainage (X. Zhou, Helmers, and Z. Qi 2013), and bimodal retention functions predict preferential flow far better than unimodal ones (Mohanty, Bowman, Hendrickx, and vanGenuchten 1997). Fully coupled surface–subsurface–tile models are recommended for detailed dynamics, with equivalent-porous-medium or seepage-node simplifications reproducing outflow almost exactly for watershed integration (De Schepper, Therrien, Refsgaard, and A. Hansen 2015; Boico et al. 2022; Xian et al. 2017).

11.3 Analytical drain-spacing solutions and pipe-scale hydraulics

Unsteady-state (transient) spacing equations yield markedly narrower spacings than steady-state equations and are preferred under variable rainfall—about 30 m versus 55 m in Iranian paddies—with the appropriate transient model depending on rainfall duration and design criterion (OOSTERBAAN 1988; Darzi-Naftchally, Mirlatifi, and Asgari 2014; Moshayedi et al. 2020; Pali, Katre, and Khalkho 2014; Nédélec, Ouvry, et al. 2010). Incorporating evapotranspiration and unsaturated-zone flow generally permits wider, more economical spacing: ET-aware solutions allow 9–18% increases, bare-soil evaporation 23–74%, and unsaturated horizontal flow up to 30% for coarse soils near a shallow impervious layer (NIKAM et al. 1992; S. Singh and Jaiswal 2010; Upadhyaya and Chauhan 2000; Abdel-Fattah, Bazaraa, and Bakr 2005; Yousfi, Mechergui, and Ritzema 2014). Bi-level and sloping-aquifer solutions show asymmetric water tables and slightly to moderately larger spacings, and dual-depth systems always remove water more effectively than single-depth ones (A. Verma, S. Gupta, K. Singh, et al. 1998; Kankarej and Upadhyaya 2023; Upadhyaya 2013; Hartani, Zimmer,

and Lesaffre 2001; Kirkham, vanderPloeg, and Horton 1997). Analytical solutions for multi-layered and ponded systems have been verified against experiments and MODFLOW, establishing strong three-dimensionality and sensitivity to anisotropy near drains, while flat-land theory remains applicable on slopes up to about 0.15 m m^{-1} (Chakrabarti and Barua 2020; Sarmah and Barua 2015; SHARMA et al. 1991; Afruzi, Nazemi, and Sadraddini 2014; Lin and Hsieh 2025; CHU 1977; FIPPS and SKAGGS 1989; LORRE, LESAFFRE, and SKAGGS 1994); no single linear-diffusion model, however, can describe all stages of drawdown (Kacimov 2000). The most complete descriptions solve the Richards equation, but corrected Boussinesq approximations remain useful for design when a variable drainable porosity or effective specific yield is used (TANG and SKAGGS 1977; Xin, Dan, et al. 2016b; VANDELEUR 1959; R. SINGH and RAI 1980; Bouarfa and Zimmer 2000). Single-tile solutions using a variable region of influence eliminate the spacing parameter for irregular systems (Cooke, Badiger, and García 2001), and space-fractional and inverse formulations improve fits in heterogeneous soils and circumvent the awkward drainable-porosity parameter (Fath-Allah 2025; Mehdinejadiani, Naseri, et al. 2013; Mehdinejadiani and Fathi 2020; Mehdinejadiani, Fathi, and Khodaverdiloo 2022; Samani, Fathi, and Homaee 2007; Chahar and Vadodaria 2010; K. SINGH et al. 1992; KUMAR, S. GUPTA, and RAM 1994; Onyejekwe 1999; Bahceci 2008).

At the pipe scale, entrance resistance and effective radius are now quantified across perforation and envelope geometries: 3–4-row pipes averaged 0.6 cm effective radius and 8-row reached 1.9 cm, while a knitted sock raised it to 5.7–6.0 cm and nearly eliminated entrance head loss, increasing perforation rows being the most efficient improvement (Ghane 2022; Ghane, Dialameh, et al. 2022). Numerical and experimental work shows rectangular slots are more hydraulically advantageous and structurally safer than circular holes, and screen envelopes reduce resistance and prevent rapid clogging (Gaj and Madramootoo 2020; Oyarce, Gurovich, and Duarte 2017; Gaj and Madramootoo 2021; Stuyt and Dierickx 2006). Within-conduit flow is a spatially varied flow problem in which Manning's equation alone is inadequate, local head loss is non-negligible, and resistance rises 40–50% over gradually varied flow (Bayat, Kouchakzadeh, and Azimi 2011; Graber 2004; Kouchakzadeh et al. 2011; Nédélec and Gay 2008; Z. Wu et al. 2022; A. Gupta et al. 2019).

11.4 Watershed representation, sensitivity, calibration, uncertainty, and evaluation

How drains and subsurface heterogeneity are conceptualised strongly controls simulated catchment behaviour: two grid conceptualisations of a Danish catchment produced drain contributions of 13% versus 31% despite both matching total streamflow within 0.1% (Karan et al. 2021). Seepage-node representations are faster and better suited to unknown drain locations (Boico et al. 2022), distributed HydroGeoSphere reproduced flashy heads better than MIKE SHE (De Schepper, Therrien, Refsgaard, X. He, et al. 2017), machine learning links streamflow signatures to simulated drain fraction (Schneider, Stisen, and Hojberg 2022), embedding drainage in integrated models reproduced streamflow without site-specific calibration (Rathore, Svyatsky, et al. 2024; Muma, Rousseau, and Gumiere 2016), and equivalent-medium representations match explicit tile simulations within 2% (Thomas et al. 2016); standard SWAT under-represents subsurface drainage without explicit adaptation (Manguerra and Engel 1998; Golmohammadi et al. 2016; Rahman et al. 2014). For regional assessment, MODFLOW matched observed heads closely, and new Hooghoudt-based subsurface-drainage packages markedly outperform the conventional DRN package (RRMSE $< 5\%$) by calibrating only conductivity (Mirlas 2012; Ghorbani et al. 2016; Y. Yang, Y. Zhu, et al. 2022; X. Han et al. 2025; Y. Qian, X. Han, Y. Zhu, W. Yang, et al. 2025).

Formal analyses converge on saturated hydraulic conductivity and evapotranspiration estimation as the dominant sources of sensitivity and uncertainty, with van Genuchten shape parameters,

drain spacing, and depth also prominent, and parameter uncertainty propagating into predicted salinity and yield (Nabi-Sichani and Sepaskhah 2012; Malakshahi, Darzi-Naftchali, and Mohseni 2020; X. Wang, Frankenberger, and Kladivko 2006; Sedaghatdoost, Ebrahimian, and Liaghat 2018; Timalsina et al. 2023; Alzraiee, Gates, and Garcia 2013). Objective-function design strongly affects identifiability: genetic algorithms generally outperform PEST, combining discharge with water-table depth already gives excellent results, and variational data assimilation cut a five-year cumulative-volume discrepancy below 1% (Sedaghatdoost, Ebrahimian, and Liaghat 2019; Chelil, Oubanas, et al. 2022; Schoups et al. 2005). The review literature concludes no universal model exists and that automatic calibration is preferable because it supports uncertainty estimation (A. Singh 2019). Crucially, evaluation choices are themselves substantive: the Kling–Gupta efficiency consistently outperformed and could reverse rankings relative to Nash–Sutcliffe, and regression-based tests exposed gross over-prediction invisible to R^2 (Bruhn et al. 2025; STONE 1995; SAVABI 1995; Wessel et al. 2022). Measurement protocols matter equally: calibrated V-notch and compound weirs, flow meters, and flow-proportional sampling minimise bias in discharge and load estimates that otherwise propagates into nutrient budgets (L. Christianson, R. Christianson, Lipka, et al. 2019; Bin Shokrana and Ghane 2021; Napierala 2023; Katuwal et al. 2024; Enamorado et al. 2007; Dialameh and Ghane 2022; Young, M. Han, and Peschel 2023).

11.5 Data-driven, remote-sensing, and geostatistical methods

Data-driven methods frequently match or outperform process models for the specific outputs on which they are trained while requiring less data and computation. Neural networks predicted real-time water-table control and saturated conductivity more accurately than pedotransfer functions (the ANN prediction of saturated conductivity reaching R^2 up to 0.96), random forests outperformed linear regression for agronomic prediction, and an Attention U-Net surrogate reached test R^2 of 0.997–1 for head, drainage volume, and salt discharge while drastically reducing computation (C. Yang, Tan, and Prasher 2000; Ellafi, Deeks, and Simmons 2021; Djiemon, Gasser, and Gallichand 2019; Lei et al. 2026; Sarangi et al. 2006; Bakhsh and Kanwar 2005; Bakhsh, Kanwar, and Malone 2007). Parsimonious surrogates (Cubist, SVM, NIT-DRAIN) can match process models for flow while training far faster, though process models remain preferable for climate extrapolation (V. Sharma et al. 2003; Negm, Youssef, Chescheir, et al. 2016; Frederiksen et al. 2023; Chelil, Henine, et al. 2022; Z. Li, Z. Qi, J. Xu, et al. 2025). Remote sensing combined with deep learning now maps drained extent at regional to national scales with accuracies commonly exceeding 85–96% and totals within a few percent of census estimates: AgTile-US reported 22.34 Mha (-0.35% versus USDA) and SEETileDrain achieved 0.96 testing accuracy (Valayamkunnath et al. 2020; Wan et al. 2024; Moller et al. 2018), while convolutional segmentation networks reliably delineate visible drain lines and outperform classical edge detectors (Redoloza et al. 2023; Song, Woo, and Yan 2021; Woo, Ji, and Song 2023; Naz, Ale, and Bowling 2009). Ancillary sensing infers drainage-relevant surface conditions through SAR, Landsat SWIR, and vegetation indices (F. Meunier, J. Meunier, and Cavayas 2001; Rijal, X. Zhang, and X. Jia 2013; M. Xu, Lacey, and Armstrong 2018; Khanal et al. 2020). Finally, kriging-family methods are consistently the most reliable interpolators for the spatially variable soil properties governing drain spacing: in the Nile Delta, conductivity was isotropic with a 5.6 km range and kriged values were recommended as design inputs, while accounting for variability widened spacing about 5% and saved substantial pipe, and neglecting K_{sat} variability could bias design spacing by -27% to $+3\%$ (GALLICHAND, PRASHER, et al. 1991; GALLICHAND, MARCOTTE, et al. 1992; GALLICHAND and MARCOTTE 1993; Moustafa and Yomota 1998a; Moustafa 2000; Moustafa and Yomota 1998b; Akis 2015; Sepaskhah and Ataee 2004).

11.6 Mapping drainage infrastructure and quantifying extent

Because buried networks are frequently undocumented, a substantial body of work establishes that non-invasive remote sensing and near-surface geophysics can detect and map subsurface drainage, with a consistent conclusion that no single method or spectral band is universally superior and that detection depends strongly on soil moisture, soil composition, drain depth and diameter, and survey timing relative to rainfall and field operations (Carlsen et al. 2024; Wienken and Grenzdörffer 2024; Koganti, Ghane, et al. 2021). In the largest multi-site assessment (29 sites), UAV visible-color imagery detected drain lines at 48% of sites, multispectral at 59%, and thermal infrared (TIR) at 69%, so acquiring all three bands is best practice (Allred, Martinez, Fessehazion, Rouse, Williamson, et al. 2020). TIR is the most consistently effective single band, detecting roughly 60% of infrastructure even under dry conditions when visible imagery failed, though it exploits apparent temperature differences below 1.0 °C, requires careful timing (optimally within 12 h of ∼2.5 mm rain and late morning to early afternoon), and can fail under low contrast, where RGB imagery guides interpretation (Allred, Eash, et al. 2018; Woo, Song, and P. Kumar 2019; Allred, Martinez, Fessehazion, Rouse, Koganti, et al. 2021; Allred, Martinez, Khanal, et al. 2022; Kratt et al. 2020). Aerial visible and multispectral photography likewise reveals functional drains through soil and vegetation contrasts and has repeatedly exposed systems absent from official registries (Tlapáková, Zaloudík, and Kolejka 2017). Ground-penetrating radar senses pipes directly, with 250–1000 MHz antennas effective, antenna orientation interacting strongly with soil moisture, and RTK/GNSS-positioned surveys resolving system type in only four to six hours without dense grids; 3D-GPR succeeds where soil electrical conductivity is low enough to limit attenuation (Allred, Redman, et al. 2005; Allred 2013; Allred, Wishart, et al. 2018; Koganti, Van de Vijver, et al. 2020). Geoelectrical methods coupling time-lapse ERT with injected water or pressurised air locate networks by induced resistivity change (Jouen et al. 2018; Henine, Tournebize, et al. 2021). The best-supported strategy is combining methods with complementary sensitivities—particularly UAV RGB/TIR imagery with GPR, which adds pipe depth and orientation—since each has distinct favourable conditions and linear farm features are otherwise easily confused with drain lines (Koganti, Ghane, et al. 2021; Carlsen et al. 2024; Wienken and Grenzdörffer 2024; Allred, Martinez, Fessehazion, Rouse, Williamson, et al. 2020).

Beyond field scale, machine learning and hydrologic filtering extend mapping to watersheds and regions: satellite big data with random forests map potential drainage, and combining remote sensing, decision-tree classification, and groundwater-table information improves watershed delineation (Cho et al. 2019; Yimer et al. 2023). Once layouts are known, GIS/GPS and DEM-based tools support design and management; a suite of QGIS plugins tested in Illinois streamlined identifying landscape features, producing topologically sound flow networks, and optimising pipe depth and size in under ten minutes versus over two hours for proprietary software (Shedekar and Brown 2018; Falasy and Cooke 2025a; Falasy and Cooke 2024; Falasy and Cooke 2025b). Quantifying extent frames the significance of drainage effects: regional mapping estimated about 25% of Germany drained (30–40% on hydromorphic soils), U.S. datasets diverge at watershed scale so best estimates remain locally conditioned, and this extent rests on validated salt- and water-balance models, empirical effluent-quality relationships, and geophysical (EMI/apparent-EC) monitoring that together allow design principles to transfer with confidence to new sites (Hirt, Meyer, and Hammann 2005; Hirt and Volk 2011; Jame et al. 2022; Y. Chen et al. 2025; F. Tian et al. 2023; Bahçeci, Dinç, et al. 2006; R. Skaggs, Youssef, and Chescheir 2006; Narjary et al. 2024; Qin et al. 2025).

12 Conclusions

Across observational, tracer, geophysical, experimental, and modelling lines of evidence, a coherent and strongly supported picture of agricultural subsurface drainage now emerges. Its foundation is hydrological: tile networks are the master control on the water balance of drained land. By intercepting water that would otherwise recharge shallow groundwater or leave as surface runoff and routing it rapidly to streams, drains commonly carry 40–65% of streamflow in humid settings and dominate field-scale water export (Schilling, Gassman, et al. 2019). This repartitioning reshapes the flow regime in consistent ways—elevating low flows and the baseflow fraction, exerting only modest and scale-dependent effects on the largest peaks, shortening groundwater travel times, and homogenising catchment response so that drained basins behave far more uniformly and scale-invariantly than their undrained counterparts. Relative to undrained or surface-drained land, improved subsurface drainage generally reduces surface runoff, sediment, and phosphorus even as it increases nitrate delivery. Because this hydrology governs everything downstream, the physical behaviour of drainage is the necessary starting point for understanding its water-quality and agronomic consequences.

That hydrological primacy is clearest for nitrate, the dominant nitrogen species leaving drained cropland and the best-quantified consequence of drainage. The subsurface pathway carries the great majority of nitrogen loss; concentrations routinely approach or exceed the 10 mg L^{-1} drinking-water limit; and loads are governed first by drainage discharge, so export is strongly seasonal, concentrating in the non-growing season and in a few high-flow episodes (Kladivko and Bowling 2021; Speir et al. 2021). Concentration–discharge behaviour is broadly chemostatic, and the nitrate is largely organic-derived and delivered rapidly through macropores, with denitrification providing real but only partial attenuation. Phosphorus tells a parallel but distinct story: although it moves in far smaller mass, the long-standing assumption of a filtered "clean" subsurface pathway has been overturned. A tiny, hydraulically active volume of biopores and cracks delivers particulate phosphorus and sediment to drains within hours, so tiles frequently dominate edge-of-field phosphorus loads despite lower concentrations than surface runoff, with dissolved losses tightly coupled to soil phosphorus status and legacy phosphorus sustaining loads at watershed scale (Pease, King, et al. 2018; Sharpley 1995).

Against these losses the evidence supports a coherent, tiered hierarchy of mitigation with quantified effects. In-field, nitrogen-rate optimisation toward the economic optimum, cover crops—especially cereal rye, clustering around 40–60% nitrate-load reduction (Malone, Obrycki, et al. 2018; Malone, Jaynes, Kaspar, et al. 2014; Malone, Radke, et al. 2023)—diversified rotations, and perennials act in predictable directions, while tillage is a weak and inconsistent lever. At the design scale, spacing and depth trade drainage volume against loss, and controlled drainage and drainage water recycling reduce loads largely in proportion to the flow they retain, delivering water-quality benefits together with yield resilience. At the edge of field, constructed wetlands, denitrifying woodchip bioreactors, reactive phosphorus media, and saturated buffers intercept loads through well-characterised controls—residence time, temperature, loading, and pH (L. Christianson, R. Christianson, Helmers, et al. 2013; Jaynes, Kaspar, et al. 2008). Two cross-cutting lessons recur: these practices redistribute nitrogen and can trade dissolved for particulate phosphorus or one loss pathway for another, so single-pathway evaluation misleads; and their benefits are scale-attenuated, diluted between the tile and the watershed outlet. Meeting downstream targets therefore requires stacking in-field and edge-of-field measures, and traditional surface-water practices that fail to intercept subsurface flow are largely ineffective over drained land. The same preferential-flow and hydrologic controls extend to pesticides, manure-borne microbes and resistance genes, trace elements, and dissolved greenhouse gases, where transient concentrations can be high even as mass

losses stay small and management often matters more than drainage itself.

Alongside these environmental costs, drainage is an established and valuable production technology. It reliably raises crop yields—most strongly on poorly drained soils and in wet years—leaches salts to reclaim saline and waterlogged land, and enables safe reuse of saline and brackish water where salt balance is maintained. Controlled variants add further yield and water-use benefits, and formal appraisals across many regions confirm favourable financial and social returns, with careful design and mechanisation substantially reducing cost. Underpinning the whole synthesis is a mature, validated toolkit: process models such as DRAINMOD reproduce drainage flow and nitrate reliably after calibration, with skill improving from daily to annual scale ; the physics of saturated–unsaturated and preferential flow is well understood; analytical spacing solutions, data-driven surrogates, and remote sensing complement one another; and geophysical and machine-learning methods now map buried networks and drained extent at field to national scales.

The strength of these conclusions rests on their convergence. Independent methods, disparate climates, and separate scales point to the same mechanisms and comparable magnitudes, so what is known about tile drainage is known with some confidence. The unifying message is that drainage is fundamentally a hydrological intervention: it sets the pathways, timing, and volume of water movement, and water-quality outcomes, agronomic gains, and the effectiveness of every management practice follow from that control. Recognising drainage water quality as a design criterion alongside production, and combining complementary practices across the field-to-watershed continuum, is the consolidated implication of the evidence assembled here.

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